

**MODULE 1:
INTRODUCTION TO
COMMODITIES AND
COMMODITY DERIVATIVES**

LEARNING OUTCOMES

- 1.a.** Compare characteristics of commodity sectors
- 1.b.** Compare the life cycle of commodity sectors from production through trading or consumption
- 1.c.** Contrast the valuation of commodities with the valuation of equities and bonds
- 1.d.** Describe types of participants in commodity futures markets
- 1.e.** Analyze the relationship between spot prices and futures prices in markets in contango and markets in backwardation
- 1.f.** Compare theories of commodity futures returns
- 1.g.** Describe, calculate, and interpret the components of total return for a fully collateralized commodity futures contract

LEARNING OUTCOMES

- 1.h.** Contrast roll return in markets in contango and markets in backwardation
- 1.i.** Describe how commodity swaps are used to obtain or modify exposure to commodities
- 1.j.** Describe how the construction of commodity indexes affects index returns

[LOS 1.a] *Compare* characteristics of commodity sectors

1.

How commodities acquire their value?

Commodities are an asset class inherently different from traditional financial assets, therefore, they are valued differently.

Traditional financial assets

When the firm generates earnings, shareholders receive their per share value.

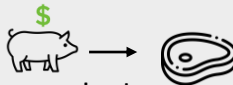


Commodities

Commodities' value derives from either their use as consumables or as inputs to the production of goods and services.



Consumables



Inputs

[LOS 1.a] Compare characteristics of commodity sectors

2. Investing in commodity

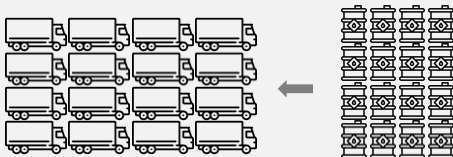
Commodities are traded in **physical (spot) markets** and in **futures and forward markets**.

Trading commodity in physical (spot) markets

- Perform **directly** between buyer and seller.
- Prices reflect **current supply and demand** conditions.
- **Involve** the physical transfer of goods.

Example: ABC Transport co. have 1000 new trucks. To fuel its trucks immediately, ABC Transport co. have to buy 1000 barrels of oil on the spot market.

ABC Transport
company



[LOS 1.a] Compare characteristics of commodity sectors

2. Investing in commodity

Commodities are traded in **physical (spot) markets** and in **futures and forward markets**.

Trading commodity in futures and forward markets

- Take place in **financial exchanges**.
- Prices established in the **contracts** on a **future date of delivery**.
- **Can be cash settled** or require physical delivery.

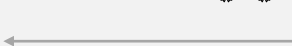
Example: Now all the truck is fully fueled. However, ABC's manager afraid that the oil price will increase in the future, when the trucks need to be fueled again. They can buy futures contracts to lock in the oil price at the present for future delivery.

ABC Transport company



Oil
futures
contracts

Futures
exchange



[LOS 1.a] *Compare* characteristics of commodity sectors

2.

Fundamental Analysis

Fundamental analysis is the process applied to predict shocks that affect supply and demand of commodities, thus affect commodity prices.

Inputs/tools used in fundamental analysis

Announcements

Forecasts produce by government or private agencies
→ used to infer demand

Component analysis

Some commodities can be broken down into components. Analysis of demand and supply forecasts of components can fine-tune the forecasts for the aggregate.

Timing issues

Incorporating any seasonality and previously observed logistical issues can refine estimates of demand and supply.

Macro

Macroeconomic factors such as inflation may trigger an increase in demand for commodities.

[LOS 1.a] *Compare* characteristics of commodity sectors

3.

Commodity Sectors

- Commodities can be defined and separated into sectors.
- The purpose of the segmentation is to isolate the characteristics that are important in supply and demand analysis in each group.
- The segmentation we are discussing is the basis for Bloomberg Commodity Index: *energy, grains, industrial (base) metals, livestock, precious metals, and softs (cash crops)*.

Sector analysis framework

Sector name: What it is and example of primary commodities.

Primary influences

Stocks: potential production or demand attempts to set boundaries around what is actually produced or wanted

Flows: considers the utilization of that stock of raw material

[LOS 1.a] *Compare* characteristics of commodity sectors

3.

Commodity Sectors

3.1. Energy

Fuel transportation, industrial production, and electrical generation.

Primary commodities

Crude oil, natural gas, coal, and refined products, such as gasoline and heating oil.

Primary influences

Stocks: Discovery and depletion of new fields, economic and political costs/certainty of access to those fields, refinery technology and maintenance, power plant type and construction, economic (GDP) size

Flows: Pipeline and tanker reliability, seasonality (summer/winter), adverse weather (cold, hurricanes), automobile/truck sales, geopolitical instability, environmental requirements, economic (GDP) growth

[LOS 1.a] *Compare* characteristics of commodity sectors

3.

Commodity Sectors

3.2. Grains

Provide human and animal sustenance but also can be distilled into fuel.

Primary commodities

Corn, soy, wheat, and rice.

Primary influences

Stocks: Arable farmland, storage/ port facilities (infrastructure), human and animal population size.

Flows: Weather (moisture, temperature), disease, consumer preferences, genetic modification, biofuel substitution, population growth

[LOS 1.a] *Compare* characteristics of commodity sectors

3.

Commodity Sectors

3.3. Industrial/Base Metals

Materials for durable consumer goods, industry, and construction.

**Primary
commodities**

Copper, aluminum, nickel, zinc, lead, tin, and iron

**Primary
influences**

Stocks: Mined acreage, smelter capacity, economic (GDP) stage of industrial/consumer development.

Flows: Government industrial and environmental policies, economic (GDP) growth, automobile/truck sales, infrastructure investment.

[LOS 1.a] *Compare* characteristics of commodity sectors

3.

Commodity Sectors

3.4. Livestock

Animals raised for human consumption.

Primary commodities

Hogs, cattle, sheep, and poultry

Primary influences

Stocks: Herd size, processing plant capacity, consumer preferences, feed availability/cost.

Flows: Speed of maturation to slaughter weight, economic (GDP) growth/consumer income, disease, adverse weather

[LOS 1.a] *Compare* characteristics of commodity sectors

3.

Commodity Sectors

3.5. Precious Metals

Certain metals that act as monetary stores of value (as well as industrial uses).

**Primary
commodities**

Gold, silver, and platinum.

**Primary
influences**

Stocks: Mined acreage,
smelter capacity, fiat
money supply/banking
development

Flows: Central bank
monetary policy,
geopolitics, economic
(GDP) growth.

[LOS 1.a] *Compare* characteristics of commodity sectors

3.

Commodity Sectors

3.6. Softs (Cash Crops)

Crops sold for income—as opposed to consumed for subsistence—and often originally seen as luxuries.

**Primary
commodities**

Cotton, cocoa, sugar, and coffee.

**Primary
influences**

Stocks: Arable farmland, storage/ port facilities (infrastructure), economic (GDP) size.

Flows: Weather (moisture, temperature), disease, consumer preferences, biofuel substitution, economic (GDP) growth/consumer income.

[LOS 1.b] *Compare* the life cycle of commodity sectors from production through trading or consumption

1. Life cycle of commodity sectors

Definition

The commodity production life cycle reflects and amplifies the changes in storage, weather, and political/economic events that shift supply and demand.

Timing/seasonality is an overlay on top of the underlying supply/demand factors.

Timing/seasonality

Short life cycle

Rapid adjustment to outside events

Long life cycle

Limits the ability of supply or demand to react to new conditions

Valuation of commodity

Shape of the supply and demand curves

Price elasticities of demand and supply

[LOS 1.b] Compare the life cycle of commodity sectors from production through trading or consumption

2. Energy life cycle

Energy commodity sector is an example showing that there can be drastic differences of life cycle of commodities in the same sector. The life cycle of two primary energy commodities: crude oil and natural gas is shown.

Crude oil life cycle	Extraction	Storage	Refining	Consumption
Description	Locate potential oil fields. A hole is dug to extract oil.	Crude oil is stored and wait to be refined.	Boiled off, and cool down then collected in each level of heat.	Distilled products are used by the end consumer.
Collected products	Crude oil		Gasoline, diesel oil, asphalt (side product).	
Natural gas life cycle	Extraction	Consumption		
Description	Same process with crude oil	Can be consumed in its natural form → No refining is required		

[LOS 1.b] *Compare* the life cycle of commodity sectors from production through trading or consumption

2. >> Energy life cycle

Significant costs in the life cycle of energy:

- Refineries are extraordinarily expensive to build, depending on the processes required to purify and distill the oil.
- Pipelines are also very costly to construct, but often destroyed or cut off.
- Costs and risks of oil exploration.

[LOS 1.b] Compare the life cycle of commodity sectors from production through trading or consumption

3. Industrial/Precious Metals Life Cycle

The life cycle of precious and industrial metal is very flexible, because the ore and the finished products can be stored for a very long time, given the resistance to spoilage of metals.

Metal life cycle	Extracting	Preparing	Smelting	Storage/ Logistic
Description	Ore is removed via a mine or open pit.	Ore is then ground into powder and concentrated to roughly 25% purity.	The purified ore is heated, and more impurities are removed as slag.	Purified metal is stored in a warehouse until shipped to end user
Collected products	Ore (~2% metal content)	Purified ore (25% metal content)	Purified metal (99.9% metal content)	

The process adds value by increasing purity

[LOS 1.b] *Compare* the life cycle of commodity sectors from production through trading or consumption

3. Industrial/Precious Metals Life Cycle

Economies of scale is created in the smelter and ore processing plants (where the purity increased most). These are huge facilities, and marginal costs decline substantially with both the scale of the facility and its utilization.



Smelter once ignited will operate on a very large scale.



Costs per unit get cheaper

Overproduction occurs: It's difficult for suppliers to cut back production or halt it entirely. Overproduction often continues until smaller or financially weaker competitors are forced to shut down.

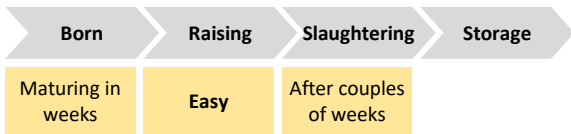
[LOS 1.b] Compare the life cycle of commodity sectors from production through trading or consumption

4. Livestock Life Cycle

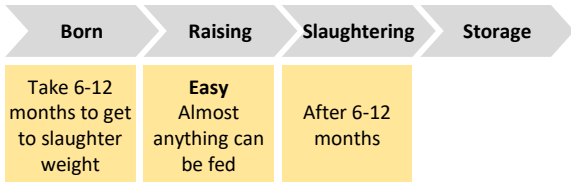
There is fluctuation in the availability of animals ready for slaughter due to external condition that affects weight gain. In the sector, there are differences between the cycle of animals.



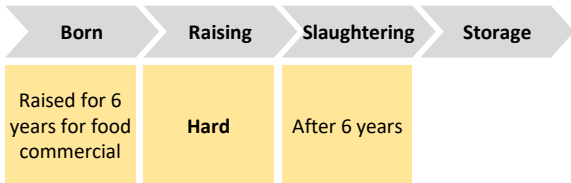
Poultry



Piglet



Cow



[LOS 1.b] Compare the life cycle of commodity sectors from production through trading or consumption

4.

Livestock Life Cycle



Advances in
cryogenics
technologies



Worldwide
delivery now
possible



Development of
middle-class
consumers



Increasing
purchase of meat
in daily diet

Increasing investment in the livestock and
meatpacking industries

[LOS 1.b] Compare the life cycle of commodity sectors from production through trading or consumption

5.

Grains Life Cycle

Grains life cycle	Planting	Growth	Formation	Harvest
Description	Placing the seeds in the ground after preparation/fertilization work	The emerging of the seedling to full height	The food grain is created by the plant.	The collection of the grain by the farmer

- Because demand for grains is year-round, they are regularly stored in silos and warehouses globally.
- Poor hygienic standards and logistics can result in a substantial loss of value to grains due to mold or insect/animal infestation.
- Farmers and consumers can trade futures to hedge their exposure to the crop.
- The contract delivery months reflect the different times of the growing cycle outlined earlier.

[LOS 1.b] Compare the life cycle of commodity sectors from production through trading or consumption

6. Softs Life Cycle

Coffee life cycle	Harvesting	Processing	Packing	Storage/shipment
Description	The coffee cherries are picked	The husk and fruit are removed, and the remaining bean dried	The “green” beans are then hulled, sorted, and bagged for their final markets	- Transported by ships to foreign buyers. - Roasted and ships to the local retailers.

The futures contracts associated with coffee are for the unroasted or “green” beans.



Afraid that prices will
decrease

Farmers and distributors

Want to hedge the sales price of production

Sell futures contracts



Afraid that prices will
increase

Coffee roasters

Want to hedge coffee bean purchase costs

Buy futures contracts

[LOS 1.c] *Contrast* the valuation of commodities with the valuation of equities and bonds

How commodities and equity and bonds are different

Equities and bonds

Represent **financial assets**

Are claims on the economic output of a business, a government, or an individual.

Periodic income

Settlement of cash payment

Valuation based on estimation of future profitability and cash flows

Commodities

Almost always **physical assets** except for electricity and weather.

Realize cash flows through purchase and sale.

Incur transportation and storage costs.

Actual delivery or settlement of cash payment

Valuation based on a discounted forecast of future possible prices

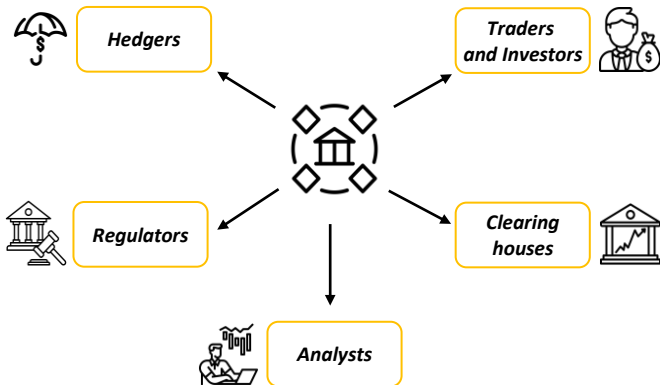
[LOS 1.d] *Describe* types of participants in commodity futures markets

1.

Overview of participants in commodity futures markets

Public commodity markets are structured as ***futures markets*** – a central exchange where participants trade standardized contracts to make and take delivery at a specified place at a specified time.

Futures market participants



[LOS 1.d] *Describe* types of participants in commodity futures markets

2.

Commodity hedgers

Definition

Who trade in the markets to hedge their exposures related to a commodity.

- Tend to be knowledgeable market participants
- May not be accurate predictors of the future supply and demand for their products.



Want to buy wheat for future delivery

Locked in a price

Drought (supply ↓)

Rising price

Benefits by buying at lower price

Beneficial weather (supply ↑)

Declining price

Losses by buying at higher price

Mitigate risk by hedging

[LOS 1.d] *Describe* types of participants in commodity futures markets

2.

Commodity hedgers

Examples of Hedging and Speculating Positions

	Long Position	Short Position
Hedging	<i>Food manufacturer</i> seeking to hedge the price of corn needed for snack chips	<i>Farmer</i> seeking to hedge the sales price of corn
Speculating	<i>Integrated oil company</i> seeking to capitalize on its knowledge of physical oil markets by making bets on future price movements	<i>Commodity trading adviser</i> (CTA) seeking to earn a profit for clients via a macro-commodity investment fund

Hedgers tend to speculate based on their perceived unique insight into market conditions and determine the amount of hedging that is appropriate.

→ Difficult to clearly distinguish between hedgers and speculators.

[LOS 1.d] *Describe* types of participants in commodity futures markets

2. Commodity hedgers

Examples of Hedging and Speculating Positions

Long Position

Short Position

Hedging

Food manufacturer seeking to hedge the price of corn needed for snack chips

Farmer seeking to hedge the sales price of corn



Afraid that prices will
increase

Afraid that prices will
decrease



Food manufacturers

Want to hedge corn purchase costs

Buy futures contracts

Farmers and distributors

Want to hedge the sales price of corn

Sell futures contracts

[LOS 1.d] *Describe* types of participants in commodity futures markets

3.

Commodity traders and investors

Definition

Who speculate on market direction or volatility and provide liquidity and price discovery for the markets in exchange for the expectation of making a profit.

Primary types of traders

Informed investors

Liquidity providers

Arbitrageurs

Hedgers

May not be accurate predictors of the future supply and demand for their product

Speculators

- Seek to out-perform the hedgers.
- May be willing to act as a liquidity providers, providing insurance service to hedgers.

- Capitalize on mispricing between the commodity and the futures price.
- May own storage facility.

[LOS 1.d] *Describe* types of participants in commodity futures markets

4.

Commodity exchanges

Definition

Exchanges (or clearing houses) set trading rules and provide the infrastructure of transmitting prices and payments.

Commodity futures markets are found throughout the world.

US markets

CME and ICE

Brazil

B3 (softs, grains, livestock)

Europe

London Metal Exchange

Asia

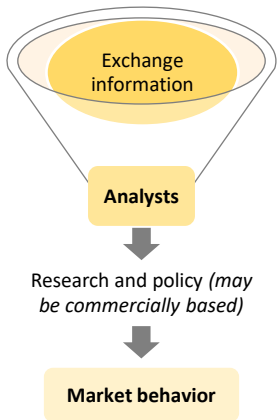
China's Dalian Commodity Exchange, Shanghai Futures Exchange and Japan's Tokyo Commodity Exchange

Indonesia (palm oil), Singapore (rubber), and Australia (energy, grains, wool) have supplementary commodity futures markets.

[LOS 1.d] *Describe* types of participants in commodity futures markets

5. Commodity market analysts

- Use information for non-trading purposes (evaluating, creating products based on commodity futures, making public policy decisions).
- Include brokers and other financial intermediaries participate in the markets but do not take a position.



Notes: Government as analysts

- Commodity prices are a key component in understanding sources of inflation
- Governments that control natural resource extraction or tax commodity extraction may want to promote or discourage investment and/or raise revenue
→ Government may also be interested in futures markets

[LOS 1.d] Describe types of participants in commodity futures markets

5.

Commodity regulators

Definition

Exist to monitor and police the markets, investigate malfeasance, and provide a venue for complaints.

Various regulatory bodies monitor the global commodity markets.

In the US

Commodity and futures regulation falls under the Commodity Futures Trading Commission (CFTC), which is a regulatory body separate from the better-known Securities and Exchange Commission.

Outside the US

Most other countries have a unified regulatory structure.

[LOS 1.e] *Analyze* the relationship between spot prices and futures prices in markets in contango and markets in backwardation

Commodity prices

Spot price

Current price to deliver a physical commodity to a specific location or purchase it and transport it away from a designated location.

Futures price

Price agreed on to deliver or receive a defined quantity (and often quality) of a commodity at a future date.

Calendar spread

The difference in price between near-term futures contract and longer-term futures contract.

Basis

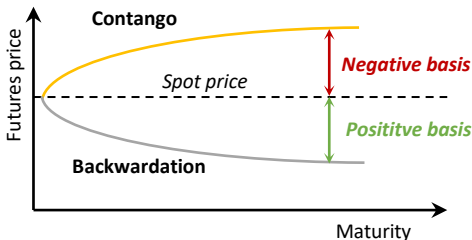
$\text{Basis} = \text{Spot price} - \text{Futures price}$

Calendar spread

$\text{Calendar spread} = \text{Near-term futures price} - \text{Longer-term futures price}$

[LOS 1.e] Analyze the relationship between spot prices and futures prices in markets in contango and markets in backwardation

Contango and backwardation markets



Contango market

- **Spot price < Futures price**
- Near-term future contract price < longer-term future contract price

Calendar spread and basis are negative

Backwardation market

- **Spot price > Futures price**
- Near-term future contract price > longer-term future contract price

Calendar spread and basis are positive

[LOS 1.f] Compare theories of commodity futures returns

Theories of returns

Insurance theory

Hedging pressure hypothesis

Theory of storage

1. Insurance theory

Statement

A commodity producer is long the physical good and thus would be motivated to sell the commodity for future delivery to hedge its sales price.



*Sell for future
delivery*



*Buy futures
contract*



*Persistent selling
puts downward
pressure*

*Compensated
for taking on
price risk*

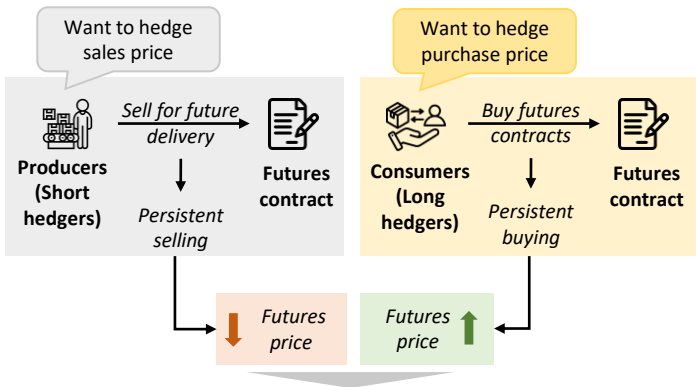
*Lower futures
price*

Backwardation market

[LOS 1.f] Compare theories of commodity futures returns

2. Hedging Pressure Hypothesis

Hedging pressure plays an important role in explaining futures returns.



Short hedgers dominates → **Backwardation** market

Long hedger dominates → **Contango** market

If the two forces of producers and consumers both seeking price protection are equal in weight, then one can envision a **flat commodity curve**.

[LOS 1.f] Compare theories of commodity futures returns

3. Theory of Storage

How the level of commodity inventories helps shape commodity futures price curves.

Demand < Supply



Demand > Supply



Convenience yield

- The benefits of holding physical inventory of the commodity.
- Acts as a buffer to potential supply disruption.

- Stock is abundant → low convenience yield
- Inventories diminish → higher convenience yield

$$\text{Futures price} = \text{Spot price} + \text{Direct storage costs} - \text{Convenience yield}$$

[LOS 1.g,h] Describe, calculate, and interpret the components of total return for a fully collateralized commodity futures contract; Contrast roll return in markets in contango and in backwardation

Total return with futures is different from total return on the physical asset

Total return

Price return



Roll return



Collateral return

Spot yield

Roll yield

Collateral yield

1.

Price return

Definition

The change in commodity futures prices of the front month contract

Price return

$$= \frac{\text{Current price} - \text{Previous price}}{\text{Previous price}}$$

[LOS 1.g,h] Describe, calculate, and interpret the components of total return for a fully collateralized commodity futures contract; Contrast roll return in markets in contango and in backwardation

Total return with futures is different from total return on the physical asset

Total return

Price return



Roll return



Collateral return

Spot yield
Roll yield
Collateral yield

2.

Roll return

Definition

Results from closing out expiring contracts and reestablishing the position in longer-dated contracts
→ **Roll the exposure**

Roll return

$$= \frac{\text{Near-term futures closing price} - \text{Further-term futures closing price}}{\text{Near-term futures closing price}} \times \% \text{ of the position being rolled}$$

[LOS 1.g,h] Describe, calculate, and interpret the components of total return for a fully collateralized commodity futures contract; Contrast roll return in markets in contango and in backwardation

Total return with futures is different from total return on the physical asset

Total return

Price return



Roll return



Collateral return

Spot yield

Roll yield

Collateral yield

2.

Roll return

Contango market

Spot price < Futures price
→ Need more money to roll the
same number of contracts

Negative roll yield

Backwardation market

Spot price > Futures price
→ Need less money to roll the
same number of contracts

Positive roll yield

[LOS 1.g,h] Describe, calculate, and interpret the components of total return for a fully collateralized commodity futures contract; Contrast roll return in markets in contango and in backwardation

Total return with futures is different from total return on the physical asset

Total return

Price return



Roll return



Collateral return

Spot yield

Roll yield

Collateral yield

3.

Collateral return

Definition

- Yield for bonds/cash used to maintain the futures position. Collateral acts as insurance for the exchange that the investor can pay for losses.
- When calculating expected return, most indices will use short-term Treasury bills.

[LOS 1.g,h] *Describe, calculate, and interpret* the components of total return for a fully collateralized commodity futures contract; *Contrast* roll return in markets in contango and in backwardation



Knowledge check

***Example:* Total Returns for Futures Contracts**

An investor has realized a 5% price return on a commodity futures contract position and a 2.5% roll return after all her contracts were rolled forward.

She had held this position for one year with collateral equal to 100% of the position at a risk-free rate of 2% per year. Calculate her total return on this position (annualized excluding leverage).

[LOS 1.g,h] *Describe, calculate, and interpret* the components of total return for a fully collateralized commodity futures contract; *Contrast* roll return in markets in contango and in backwardation



Knowledge check

***Example:* Total Returns for Futures Contracts**

Answer:

Total return on a commodity futures position is expressed as

$$\text{Total return} = \text{Price return} + \text{Roll return} + \text{Collateral return}$$

In this case, she held the contracts for one year, so the price return of 5% is an annualized figure. In addition, the roll return is also an annual 2.5%. Her collateral return equals $2\% \text{ per year} \times 100\% \text{ initial collateral investment} = 2\%$.

So, her total return (annualized) is

$$\text{Total return} = 5\% + 2.5\% + 2\% = 9.5\%.$$

[LOS 1.g,h] *Describe, calculate, and interpret* the components of total return for a fully collateralized commodity futures contract; *Contrast* roll return in markets in contango and in backwardation



Knowledge check

Example: Total Returns for Futures Contracts

An investor has a \$10,000 position in long futures contracts (for a hypothetical commodity) that he wants to roll forward. The current contracts, which are close to expiration, are valued at \$4.00 per contract, whereas the longer-term contract he wants to roll into is valued at \$2.50 per contract.

What are the transactions—in terms of buying and selling new contracts—he needs to execute in order to maintain his current exposure?

- Close out (sell) 2,500 near-term contracts and initiate (buy) 4,000 of the longer-term contracts.
- Close out (buy) 2,500 near-term contracts and initiate (sell) 4,000 of the longer-term contracts.
- Let the 2,500 near-term contracts expire and use any proceeds to purchase an additional 2,500 of the longer-term contracts.

[LOS 1.g,h] *Describe, calculate, and interpret* the components of total return for a fully collateralized commodity futures contract; *Contrast* roll return in markets in contango and in backwardation



Knowledge check

***Example:* Total Returns for Futures Contracts**

Answer:

A is correct. To roll over the same level of total exposure (\$10,000), he will need to do the following:

1. Sell

$\$10,000 / \$4.00 \text{ per contract} = 2,500 \text{ existing contracts.}$

2. Replace this position by purchasing

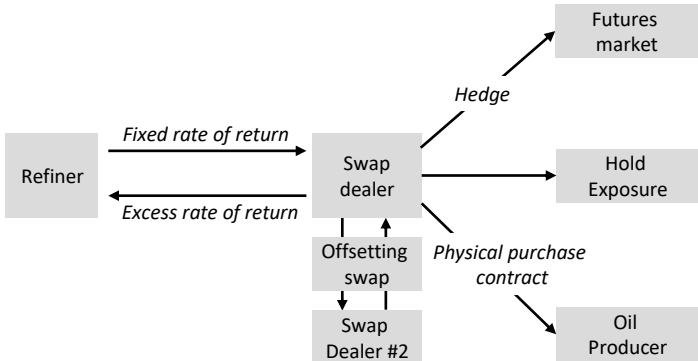
$\$10,000 / \$2.50 \text{ per contract} = 4,000 \text{ existing contracts}$

[LOS 1.i] Describe how commodity swaps are used to obtain or modify exposure to commodities

Definition

- **Commodity swap** is a legal contract involving the exchange of payments over multiple dates as determined by specified reference prices or indexes relating to commodities.
- **Series of futures contract** often forms the basis of the reference prices.

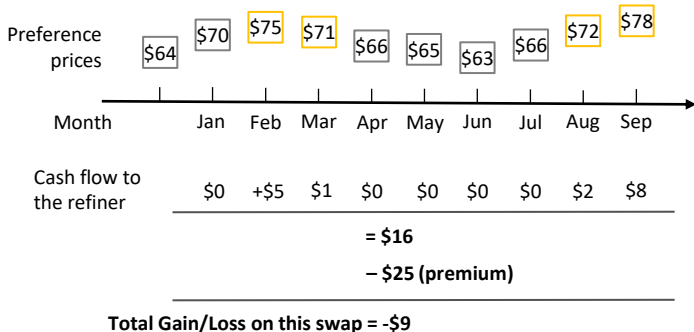
Example of an oil refiner: An oil refiner wants to hedge its oil purchases, without having to manage a large number of futures contracts
→ Entering a **swap contract** (*excess return swap*).



[LOS 1.i] *Describe* how commodity swaps are used to obtain or modify exposure to commodities

Assume we had the following scenario:

1. An oil refiner goes long a swap at the end of December that pays the amount exceeding \$70 per barrel every month-end through September.
2. The oil refiner would pay a swap counterparty a premium (in this example, \$25) for this privilege because it is effectively long a series of call options.



[LOS 1.i] Describe how commodity swaps are used to obtain or modify exposure to commodities

Total return swap

The swap buyer (the long) will receive periodic payments based on the change in the futures price of a commodity plus the return on the collateral, in return for a series of fixed payments.

Basis swap

Periodic payments are exchanged based on the values of 2 related commodity reference prices that are not perfectly correlated

Variance swaps

Two parties agree to periodically exchange payments based on the proportional **difference between an observed/actual variance** in the price levels of a commodity, and some **fixed amount of variance** established at the outset of the contract.

Volatility swaps

Relative to the volatility of a reference commodity: two sides are not speculating on how volatile prices will be versus expectations.

- A volatility seller will profit if **realized volatility is lower than expectations**
- The counterparty volatility buyer anticipates **higher than expected volatility**

[LOS 1.j] Describe how the construction of commodity indexes affects index returns

Roles of a commodity index

Used as a benchmark to evaluate broader moves in commodity pricing

Used for macro economic or forecasting purposes

Act as the basis for an investment vehicle or contract

Key characteristics that differentiate the indexes:

Breadth of coverage

Number of commodities and sectors

Relative weighting

Assigned to each component and how those weights are determined

The rolling methodology

Has a direct impact on the roll return, active or passive

Methodology and frequency of rebalancing the weights

Potential for positive rebalancing return

Governance of indexes

The process by which the rules are implemented

*Index should be **investable** (replicable with available futures contracts).*

**MODULE 2:
OVERVIEW OF TYPES OF REAL
ESTATE INVESTMENT**

LEARNING OUTCOMES

- 2.a.** Compare the characteristics, classifications, principal risks, and basic forms of public and private real estate investments
- 2.b.** Explain portfolio roles and economic value determinants of real estate investments
- 2.c.** Discuss commercial property types, including their distinctive investment characteristic
- 2.d.** Explain the due diligence process for both private and public equity real estate investment
- 2.e.** Discuss real estate investment indexes, including their construction and potential biases

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

1.

Overview of real estate investments

Real estate offers investors:

Long-term stable
income

Protection from
inflation

Low correlations with
stocks and bonds

Effective means of diversification in many portfolios

Suitable for
investors
with

Long-term
investment
horizons

The ability to
accept relatively
lower liquidity

Pension funds

Sovereign wealth funds

Insurance companies

High-net-worth individuals

**Securitized real estate ownership has historically provided smaller investors with ready access to the asset class.*

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

1. Overview of real estate investments

Risk profile of the underlying investment



High-quality properties
*Long-term leases,
low leverage*



Older properties
*Short-term leases,
high leverage*



Development property
*Long lead time,
dependence on
other parties*



2. Real Estate Investment: Basic Forms

Types of real estate classifications

Property (2.1)

Capital position (2.2)

Investment vehicle (2.3)

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

2.

Real Estate Investment: Basic Forms

2.1. Classify by property types

	Owner Occupied	For Rent (Commercial)
Residential	• Single-family homes • Apartments/condominiums* • Manufactured housing	• Single-family homes • Multi-family buildings**
Non-residential	• Office, shopping centers, • Manufacturing facilities, warehouses, agricultural • Other specialty real estate	• Office, shopping centers, • Industrial warehouses/distributions • Hotels, agricultural, other specialty real estate

* *Condominiums refer to owner-occupied units in multi-unit buildings.*

** *Multi-family properties contain individual for-rent apartments or flats.*

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

2.

Real Estate Investment: Basic Forms

2.2. Classify by capital position

Whether the investment is structured as *equity* or *debt*.

Equity investment

- Investor may be the *sole or a joint owner* of the real estate property or may *invest in securities* of a company that owns the real estate property (**REITs**)
- Higher rate of return, higher risk
- Source of return: rent income, capital appreciation

Debt investment

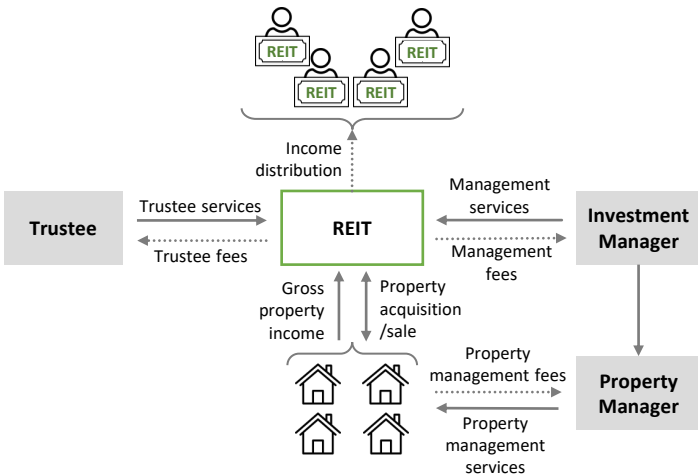
- Investors may *lend funds to the entity* acquiring the real estate property or may *invest in securities* that are based on real estate lending (**REOCs**)
- Lower rate of return, lower risk
- Source of return: promised cash flows → Similar to bonds

[LOS 2.a] Compare the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

2. Real Estate Investment: Basic Forms

2.2. Classify by capital position

Real Estate Investment Trust (REIT) Structure



[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

2.

Real Estate Investment: Basic Forms

2.3. Classify by investment vehicles

Vehicle through which the RE investment is made

Public vehicles

Public equity

- Allow the property to remain undivided while allowing investors divided ownership.
- Liquid.
- Investors need no property management expertise

Public debt

Private vehicles

Private equity

- Real estate is indivisible and illiquid → Private investments are usually larger.
- Illiquid.
- Require property management expertise.

Private debt

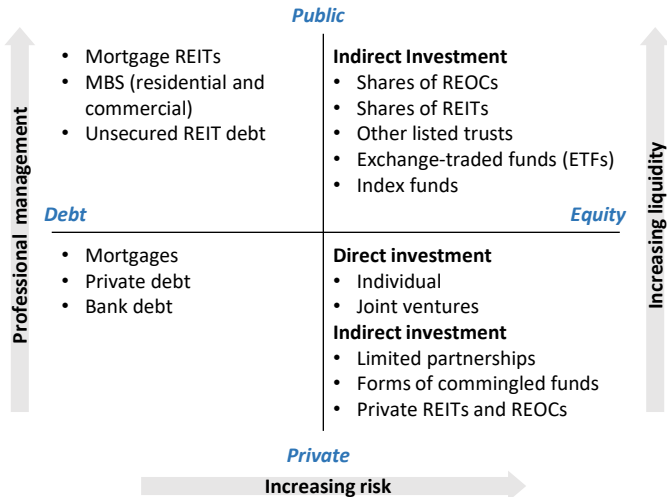
Combining the dimensions, we have four quadrants on the next slide.

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

2.

Real Estate Investment: Basic Forms

2.3. Classify by investment vehicles



[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

3. Characteristics of real estate investment

Some of the main characteristics of real estate investment that distinguish it from listed equity or fixed-income investments:

Unique Asset & Fixed Location	No two real estate properties are the same.
High Unit Value	The price of a unit of real estate property is high compared with the price of a single stock or bond.
Management Intensive	Owning real estate requires maintenance, contracting, and collecting rent.
High Transaction Costs	Buying/selling real estate is costly in time and money with multiple parties involved.
Depreciation	Deduction in the value of the building over time due to obsolescence and deterioration. This deduction has favorable tax consequences in most markets.

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

3.

Characteristics of real estate investment

Illiquidity

Real estate investments require time to sell, and the bid/ask spread is much wider than that for stocks and bonds.

Need for debt capital

Real estate investments typically require debt capital.

Price Determination

Prices are typically determined by appraisals and estimates rather than by market transactions readily visible in the market because each property is unique.

These characteristics slowed widespread investor allocations to real estate.

→ **Securitization** helped overcome some of these problems, especially investment size and illiquidity (*through REITs and REOCs*).

[LOS 2.a] Compare the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

4.

Principal risks

Sources of risk

Property demand and supply

Valuation

Property operations

Affect **income and value** of the real estate property or equity/debt investment

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

4.

Risk factors

4.1. Property demand and supply

Feature	Description
<i>Business conditions</i>	General economic conditions (GDP, employment, interest rates, and business cycle)
<i>Demographics</i>	Demographic factors (population, age, and income level)
<i>Real estate cycle</i>	Long lead times and a relatively long real estate cycle contribute to a buildup of excess supply based on the assumption of high rents.

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

4.

Risk factors

4.2. Valuation

Feature	Description
<i>Cost and availability of capital</i>	Real estate must compete with other assets for debt and equity capital. Real estate investors would be willing to pay more in a low interest rate environment.
<i>Availability of information</i>	If information on a property is limited, it is more difficult to make an informed decision on investing in that property.
<i>Lack of liquidity</i>	Real estate takes time (and money) to sell and buy.
<i>Interest rate environment</i>	The interest rate environment effects supply, demand, and prices of real estate.

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

4.

Risk factors

4.2. Valuation

Feature	Description
Management	Requires “asset management” and “property management”
Lease Provisions	Legal contracts can add to or detract from the value of the investment.
Leverage	Typically include more leverage for the equity holder than other types of investments.
ESG Considerations	Real estate values can be directly impacted by environment-, social-, and governance-related issues.
Obsolescence	Can become obsolete due to changing technology, tenant preferences, and other factors.
Ongoing market disruption	Online shopping, remote working, and other disruptions can change the face of real estate markets globally

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

4.

Risk factors



Knowledge check

Example: Commercial Real Estate Risk

An investor wants to add real estate to her portfolio to benefit from its diversifying characteristics. She decides to buy a commercial property, financing at most 30% of the asset with debt in order to avoid incurring financial risk due to interest rate changes. This strategy is most likely to:

- A. limit the risk due to leverage.
- B. mitigate the risk due to inflation.
- C. eliminate the risk due to interest rate changes.

[LOS 2.a] *Compare* the characteristics, classifications, principal risks, and basic forms of public and private real estate investments

4.

Risk factors



Knowledge check

Answer:

If less money is borrowed, there is less risk of cash flow and equity value volatility due to the use of financial leverage. → **A is correct.**

The risk related to changes in interest rates remains and cannot be eliminated, even if the investor does not borrow money. If interest rates rise:

- Investors require a higher rate of return on all assets → affect property's value.
- Affect cost of debt of the next buyer → affect resale price of the property.

→ **C is incorrect.**

There is still risk of inflation, although real estate tends to have a low amount of inflation risk. Inflation benefits fixed-rate borrowers who are able to repay debt in the future with cash that is worth less than cash borrowed today.

→ **B is incorrect.**

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

Real estate cash flow

Operating expenses

Rental income

Capital spending

Leverage

Driven by

Supply of space

Demand for space

Other economic
factors

Demand affected by

Macro factors

Individuals

Business environment

Industrial

[LOS 2.b] Explain portfolio roles and economic value determinants of real estate investments

1.

Economic drivers affecting demand

1.1. Macro factors

Macro factors **affect all** major real estate sectors, in which GDP growth is the most important single factor.

	Multi-Family	Retail	Hotel	Office	Industrial
GDP growth	✓	✓	✓	✓	✓
Population growth	✓	✓	✓	✓	✓
Job creation	✓	✓	✓	✓	✓
Wage growth	✓	✓	✓	✓	✓
Regulatory	✓	✓	✓	✓	✓
Taxes	✓	✓	✓	✓	✓

- **Job creation** → demand for office space, retail space to cater, multi-family houses, hotel room demand as both leisure and business travel
- **Wage growth** increases in disposable income → increased spending → support retail sales growth → influence rental income and occupancy

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

1.

Economic drivers affecting demand

1.2. Individual (consumer) factors

	Multi-Family	Retail	Hotel	Office	Industrial
Household formations	✓	✓			
Personal income	✓	✓	✓		
Consumer confidence	✓	✓	✓		
Consumer credit	✓	✓	✓		

- **Household formation** is one of the largest drivers of apartment demand.
- Income, wage growth, and consumer confidence → determine whether residents can afford to move to larger, higher-quality, better-located units or buy a home.

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

1.

Economic drivers affecting demand

1.3. Business environment factors

	Multi-Family	Retail	Hotel	Office	Industrial
Retail sales growth		✓			✓
Consumer spending		✓	✓		✓
Business formations	✓		✓	✓	✓
Business investment			✓	✓	✓
Business confidence			✓	✓	✓

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

1.

Economic drivers affecting demand

1.4. Industrial factors

	Multi-Family	Retail	Hotel	Office	Industrial
Industrial production					✓
Trade, transport, and logistics					✓
Changing supply routes					✓

- Warehouse distribution centers have seen increased demand from global trade in and near port cities.
- Online sales boom → store inventory closer to customers to speed delivery times

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

1.**Economic drivers affecting demand**

Knowledge check

Example 1: Economic Value Drivers

Which of the following statistics is likely to be most relevant for all of the following: office, industrial, and hotel properties?

- A. Business confidence
- B. Household formation
- C. Industrial production

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

1.

Economic drivers affecting demand



Knowledge check

Example 1: Economic Value Drivers

Which of the following statistics is likely to be most relevant for all of the following: office, industrial, and hotel properties?

- A. Business confidence
- B. Household formation
- C. Industrial production

Answer:

Business confidence is rising → companies are more likely to expand their business and engage in business travel. → **A is correct.**

Household formations increase → indirect effect on these sectors and are most relevant to the multi-family sector. → **B is incorrect.**

Changes in industrial production → less directly tied to the office and hotel markets. → **C is incorrect.**

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

1.

Economic drivers affecting demand



Knowledge check

Example 2: Economic Value Drivers

In addition to the market and property-specific analysis of occupancy, rental rate, lease expiry, and financing statistics, analysts of office properties are most likely to pay particular attention to trends in:

- A. Retail sales growth.
- B. Household formation
- C. Job creation

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

1.

Economic drivers affecting demand



Knowledge check

Example 2: Economic Value Drivers

In addition to the market and property-specific analysis of occupancy, rental rate, lease expiry, and financing statistics, analysts of office properties are most likely to pay particular attention to trends in:

- A. Retail sales growth.
- B. Household formation
- C. Job creation

Answer:

Job creation is most significant for office REITs. → **C is correct.**

Household formations are more significant for multi-family and retail REITs than for office REITs, whereas retail sales growth is more significant for shopping center/retail and industrial REITs than for office REITs. → **A and B are incorrect.**

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

1.**Economic drivers affecting demand**

Knowledge check

Example 3: Economic Value Drivers

Which of the following property sectors would be expected to experience the greatest cash flow volatility?

- A. Industrial
- B. Hotel
- C. Shopping center

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

1.

Economic drivers affecting demand



Knowledge check

Example 3: Economic Value Drivers

Which of the following property sectors would be expected to experience the greatest cash flow volatility?

- A. Industrial
- B. Hotel
- C. Shopping center

Answer:

Hotel room demand fluctuates with economic activity and business and consumer confidence; there are no long-term leases on hotel rooms to protect hotel REITs' revenue streams from changes in demand. → **B is correct.**

Industrial and shopping center REITs benefit from long-term leases on their properties and from the relatively mild dislocations between supply and demand caused by the construction of new space in these subsectors. → **A and C are incorrect.**

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

2. Real estate cycles

The supply side of the real estate economic cycle is driven by the following types of periods:

Oversupply

When occupancy and rental rates are **low**

Undersupply

When occupancy and rental rates are **high**

Property types with long development and construction periods are more prone to supply–demand dislocations for the following reasons:

1. New construction typically **commences in a booming economy**.
2. Properties already under development continue to be completed after a **recession eventually arrives** and depresses demand.
3. The large size of many facilities, complicated mixed-use properties, or convention center hotels further **exacerbate excess supply on completion**.

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

3.

Role of Real Estate in an Investment Portfolio

Core real estate investment style/strategy

Investing in high-quality, well-leased, core property types with low leverage (no more than 30% of asset value) in the largest markets with strong, diversified economies.



Conservative strategy designed to minimize real estate-specific risks.

Five reasons for investors to invest in real estate:

3.1. Current income

Investors earn current income on the property through letting, leasing, or renting the property.

Expect that market demand will be sufficient to produce net income after collecting rents and paying operating expenses.

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

3.

Role of Real Estate in an Investment Portfolio

3.2. Capital appreciation

Investors may anticipate selling properties after holding them for a period of time and realizing the capital appreciation.

Price increases contribute to an investor's total return.

3.3. Inflation hedge

Investors may expect both rents and real estate prices to rise in an inflationary environment.



Inflation ↑



Rents and prices ↑



Real rate of return may be less volatile than the nominal rate of return.

Inflation hedge

[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

3.

Role of Real Estate in an Investment Portfolio

3.4. Diversification

Real estate performance has not typically been highly correlated with the performance of other asset classes.

Adding real estate to a portfolio often **lowers the overall risk** of the portfolio relative to the expected return.

3.5. Tax benefits

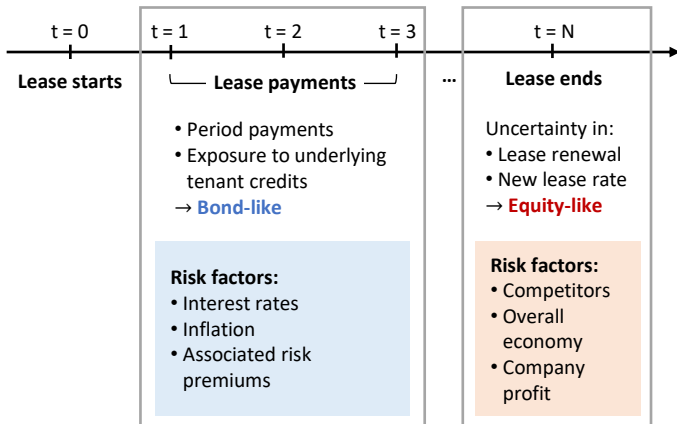
Private real estate investments may receive favorable tax treatment compared with other investments.

[LOS 2.b] Explain portfolio roles and economic value determinants of real estate investments

5.

Real Estate Risk and Return Relative to Stocks and Bonds

Structure of leases: Legal agreements requiring the tenant to make periodic payments to the space owner.

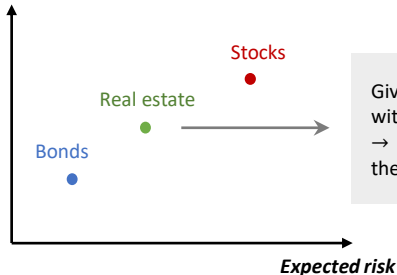


[LOS 2.b] *Explain* portfolio roles and economic value determinants of real estate investments

5. Real Estate Risk and Return Relative to Stocks and Bonds

Because of these two influences (**bond-like** and **equity-like** characteristics), core real estate is expected to have a risk and return profile that **falls between** the risk and return profiles of **large-cap stocks** and **investment-grade bonds**.

Expected Return



Given different correlations with stock and bond returns
→ Adding real estate **expands** the efficient frontier

[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

1. >> Classifications

Commercial property types include the following:

Residential properties:

Non-owner occupied,
purchased with the intent to
rent out in order to produce
income.

Single-family houses

Multi-family houses

Demand depends on **population growth**

Office buildings

Demand is dependent on **job growth**

Industrial and warehouse

Demand is dependent on **the economy**

Retail

Demand is dependent on **spending**

Hospitality

Hotels, restaurants, etc.

Others: Custom projects

Non-residential properties:

Commercial properties typically
described by their end use.

[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

2.

Investment Characteristics

Some commercial property types are more **management intensive** than others (hotels, shopping centers)

Are considered **riskier** because of the operational risks

Investors typically require a higher rate of return on these management-intensive properties

[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

2. Investment Characteristics

2.1. Risk and return profile

Risk profile for different investment type:

High-quality, well-leased **office, retail**, industrial **warehouse**, and **multi-family** properties

Low risk

Hotels with no leases and performance highly correlated to business cycle

Riskier

Hospitals and cell towers have only one type of tenant, and the facilities are not easily converted to other uses

Considered separately each

*For each property type, **location** is the critical factor in determining value.*

Best locations, modern features and functionality

Highest value

Not prime locations, slightly outdated features

Moderate value

In poor locations and have outdated features

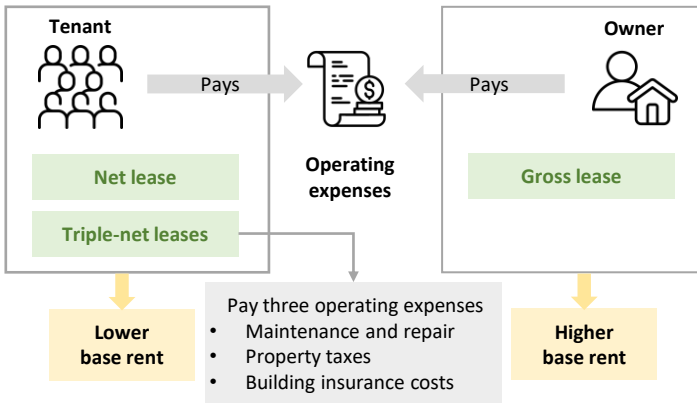
Lowest value

[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

2. Investment Characteristics

2.2. Common Types of Leases

Net lease vs. Gross lease



[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

2. Investment Characteristics

2.2. Common Types of Leases

Sale-leaseback

The company then signs a long-term lease with the buyer to continue to occupy the building and pay all expenses.

Step 1: Sale

A company sells the building it owns and occupies to a real estate investor

Step 2: Lease back

Step 3: Lease end

- Use of the property reverts to the landlord.
- The tenant is not responsible for normal property depreciation.

[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

2. Investment Characteristics



Knowledge check

Example: Net and gross lease

What is the net rent equivalent for an office building where the gross rent is \$20 per square foot and operating expenses are currently \$8 per square foot?

Answer:

For net lease:

$$\text{Rental fee} = \$20\text{psf} - \$8\text{psf} = \$12\text{psf}$$

With the gross lease, the owner bears the risk of rising operating expenses, whereas the same is true of the tenant with net leases. If expenses decline:

- Under gross lease: owner benefits through improved operating margins
- Under net lease: tenant would benefit from paying a lower amount to the landlord.

[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

2. Investment Characteristics

2.3. Lease terms

- **Medium- to long-term leases** frequently include contractual increases in rents known as rent bumps, rent step-ups, or step-up rents.
- It is common to specify step-ups tied to inflation, either annually or cumulatively after several years.

Example: A step-up of 3% of the prior-period rent every five years; occasionally an adjustment to mark rents to then-prevailing market rates.

	Long-term leases	Short-term leases
Cash flow stability	Greater	Not as good
Marking to market of rents	• Negatively affected when rates rise. • Landlords cannot raise rents on existing leases.	• Negatively affected when rates decline. • Tenants have greater ability to negotiate lower rates.

[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

2. Investment Characteristics

2.4. Investment characteristics by Property types

Office

The demand for office properties depends on

Employment growth

Typical amount of space used
per employee

Tends **to decrease over time**
as technology improve

The average length of an office building lease varies.

Lease terms may depend on

Desirability of the
property

Financial strength of
the tenant

Other terms in the
lease

[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

2. Investment Characteristics

2.4. Investment characteristics by Property types

Industrial and Warehouse

The demand for industrial and warehouse space depends on

Overall strength of the
economy and economic
growth

Import and export activity in
the economy

- Industrial leases are often long-term net leases.
- Warehouse property values have shifted along with changing trade routes.
- The retail trend has increased demand for space closer to population centers.

[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

2. Investment Characteristics

2.4. Investment characteristics by Property types

Retail

The demand depends on **trends in consumer spending**

Depends on the health of the economy, population growth, job growth, consumer confidence, and savings rates

- Lease terms vary by the size and the importance of the tenant.
- Anchor tenants may be offered extremely favorable rental terms designed to attract them to the property.
- Tenants are required to pay additional rent once their sales reach a certain level (percentage lease).

[LOS 2.c] *Discuss commercial property types, including their distinctive investment characteristics*

2. Investment Characteristics

2.4. Investment characteristics by Property types

Multi-Family

The demand for multi-family space depends on

Population growth (especially the age segment most likely to rent apartments)

Cost of renting compares with the cost of owning (ratio of home prices to rents)

Cost of renting compares with the cost of owning

Home prices

Home prices rise → more people will rent and vice versa

Interest rates

Interest rates increase cause a shift toward renting and vice versa

[LOS 2.d] *Explain* the due diligence process for both private and public equity real estate investment

Why should investors perform thorough due diligence?

Direct investment

Property value will be based on several factors: cash flow outlook, market conditions, and prices paid for recent properties.

Due diligence should include an analysis of all the cash flow drivers and liabilities.

Indirect investment

Determine whether the **share valuation** is appropriate.

Due diligence on REITs and REOCs from public filings of annual audited financial statements, quarterly reports, and investor presentations.

The purpose of due diligence is to identify potential problems that have not been disclosed by the seller that could negatively impact value.

[LOS 2.d] Explain the due diligence process for both private and public equity real estate investment

Property due diligence should include an examination of the following.

1. Market review

- Understand market trends, including local market population, job, and income growth.
- Understand expected additions to supply and space absorption rates.
- Understand tenant preferences, building amenities, market rents, and expense trends.

2. Lease and rent review

- Compare the tenant rents with market rent forecasts and lease length to determine how much rents will change when leases expire.
- Review the lease expiration timeline for all tenants.
- Analyze the history of rental payments, late payments, and any defaults for the major tenants.

[LOS 2.d] *Explain* the due diligence process for both private and public equity real estate investment

3. Review costs of re-leasing

- Review costs and incentives provided to both renewing and new tenants.
 - **Costs:** commissions paid to real estate brokers and downtime between leases.
 - **Incentives:** a period of free rent and allowances for tenants' improvements to their space
- These costs are typically not included in annual operating income but capitalized and amortized over the length of the lease.

4. Review documentation

- Review copies of bills for operating expenses.
- Review multiple years of audited financial statements.
- Look for evidence of overstated income (underspending on capital expenditures) or overstated occupancy (tenant incentives).

[LOS 2.d] *Explain* the due diligence process for both private and public equity real estate investment

5. Property inspections and service agreements

- Conduct ***environmental inspection***: to ensure there is no environmental issues.
- Conduct ***physical/engineering inspection***: to ensure there is no structural issues.
- Review ***service and maintenance agreements*** to determine whether recurring problems exist.
- Conduct a ***property survey*** to determine whether the planned physical improvements are in the boundary lines of the site.

6. Legal documentation and tax compliance

- Conduct a ***title search*** by reviewing the ownership history.
- Verify that the property is compliant with zoning ***laws***, environmental ***regulations***, etc.
- Verify that property ***taxes, insurance***, special assessments, and so on, have been paid.

If due diligence issues are discovered, the investor should adjust the valuation model accordingly and **negotiate lower price**.

[LOS 2.e] *Discuss* real estate investment indexes, including their construction and potential biases

Real estate indexes measure

Property income
performance

Property total
return

Investment fund
performance

Listed security
returns

Investors should be aware of **how the index is constructed** and the **inherent limitations** resulting from the construction method.

1.

Appraisal-Based Indexes

Insufficient transactions of the same property to rely on transactions to indicate value

Property and private real estate investment indexes often rely on appraisals to estimate values



Property A



Appraised



Property B



Appraised



Property C



Appraised



Appraisal-based indexes
provide a measure of
market movements

[LOS 2.e] *Discuss* real estate investment indexes, including their construction and potential biases

1. Appraisal-Based Indexes

The return for all the properties in the appraisal-based index is calculated as follows:

Holding period return

$$= \frac{\text{NOI} - \text{Capital expenditure} + (\text{Ending market value} - \text{Beginning market value})}{\text{Beginning market value}}$$

NOI = Net operating income

The beginning and ending market values are based on the appraisals of the properties.

This return is equivalent to a single-period internal rate of return (IRR).

Difference with measures of returns on stocks and bonds:

- Stocks and bonds: actual transaction price is typically used
- Real estates: appraised value is used

Drawback: The income component of real estate returns does not represent distributions to investors in real estate funds or REITs.

[LOS 2.e] *Discuss* real estate investment indexes, including their construction and potential biases

2. Transaction-Based Indexes

Transaction-based indexes are **based on actual transactions** rather than appraised values. However, the fact that the same property does not sell very frequently is still an issue.

Econometric techniques are used to control for the sale of different properties and to create indexes in 2 main ways.

Repeat sales index

Relies on repeat sales of the same property



Property A



Price 1



Property A



Price 2

Change in value between the two sale dates indicates how market conditions have changed over time

Hedonic index

Require only one sale



Property A



Price 1



Property B



Price 2

Separate difference in value from difference in size, age, quality of construction, and location.

[LOS 2.e] *Discuss* real estate investment indexes, including their construction and potential biases

3.

Advantages and Disadvantages of Indexes

3.1. Appraisal lags

Appraisal lags in rising and falling markets

In rising/falling market:

Transaction
prices start to
rise/fall first

Higher/lower prices
are later reflected in
comparable sales
and surveys

Captured in
appraised
values

Appraisal-based indexes lag behind the market

Infrequent Appraisals

All properties in appraisal-based index may not be appraised every quarter

A lag is caused in the index

[LOS 2.e] Discuss real estate investment indexes, including their construction and potential biases

3.

Advantages and Disadvantages of Indexes

3.2. Impact on Performance Measurement and Asset Allocation

If the purpose of the index is for comparison with other asset classes that are publicly traded, appraisal lag is an issue.

Appraisal lag tends to smooth the index (less volatility)

Underestimate
volatility of real
estate returns

Lower correlation
with other asset
classes

Overstate Sharpe
ratios

The appropriate allocation to and benefits from private real estate would likely be overestimated

[LOS 2.e] *Discuss* real estate investment indexes, including their construction and potential biases

3.

Advantages and Disadvantages of Indexes

3.3. Adjustment for Appraisal Lag

Appraisal lag can be adjusted for in two ways

“Unsmooth” the
appraisal-based index

More volatility and more
correlation with other asset
classes

Use a transaction-based index
when comparing real estate with
other asset classes

Lead appraisal-based indexes
but can be noisy (include
random elements in the
observations)

[LOS 2.e] *Discuss* real estate investment indexes, including their construction and potential biases

4.

Real Estate Security Indexes

There are a wide variety of real estate security indexes available developed by index providers, stock exchanges, and trade organizations.

Example:

- Real estate equity security indexes: Bloomberg, FTSE Russell, MSCI, Nikkei, and S&P Dow Jones.
- Real estate debt security indexes: CMBX
- Other real estate indexes: US REITs—FTSE and Nareit
- European and Asia Pacific listed real estate company indexes: FTSE, EPRA, and Nareit
- Various global real estate securities indexes: FTSE, EPRA, and Nareit

**MODULE 3:
INVESTMENTS IN REAL ESTATE
THROUGH PUBLICLY TRADED
SECURITIES**

LEARNING OUTCOMES

- 3.a.** Discuss types of publicly traded real estate securities
- 3.b.** Justify the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income
- 3.c.** Describe the use of funds from operations (FFO) and adjusted funds from operations (AFFO) in REIT valuation
- 3.d.** Calculate and interpret the value of a REIT share using the net asset value, relative value (price-to-FFO and price-to-AFFO), and discounted cash flow approaches
- 3.e.** Explain advantages and disadvantages of investing in real estate through publicly traded securities compared to private vehicles

[LOS 3.a] *Discuss* types of publicly traded real estate securities

Publicly traded real estate securities

Allow investors to gain indirect exposure to real estate equity and debt

Makes it possible for investors of all sizes to access the real estate asset class

1. >> Types of publicly traded real estate securities

Publicly traded real estate securities

Real estate investment trusts (REITs)

Real estate operating companies (REOCs)

Mortgage-backed securities (MBS)

[LOS 3.a] Discuss types of publicly traded real estate securities

1.

Types of publicly traded real estate securities

Real estate investment trusts (REITs)

- Companies that own, finance, and develop **income producing real estate**.
- Most REITs are required to distribute 90%–100% of their taxable income to shareholders → deductible dividends → **exempts REITs from CIT**.
- REITs that own real estate are called equity REITs. Those that make or invest in loans secured by real estate are categorized as mortgage REITs.

Real estate operating companies (REOCs)

Ordinary **taxable** real estate ownership companies.

Businesses are organized as REOCs if they:

- Are located in countries that do not have a tax-advantaged REIT regime in place.
- Engage to a large extent in the development of **for-sale real estate properties**.
- Offer other non-qualifying services.

Mortgage-backed securities (MBS)

Represent rights to receive cash flows from portfolios of mortgage loans, including mortgage loans on **commercial properties (CMBS)** and on **residential properties (RMBS)**.

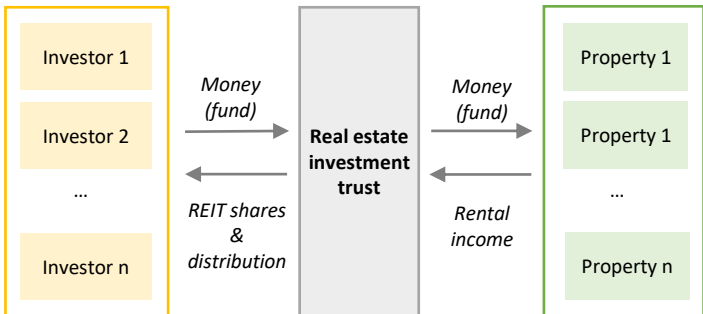
[LOS 3.a] Discuss types of publicly traded real estate securities

2. REIT Structures



Recall: Real Estate Investment Trusts

Real Estate Investment Trusts (REITs) is a type of closed-end investment company that sells shares to investors and invests the proceeds in various types of real estate and real estate mortgage.



[LOS 3.a] Discuss types of publicly traded real estate securities

2. REIT Structures

In most countries, REITs are required to

Distribute 90%–
100% of their
otherwise taxable
earnings

Invest at least 75% of
their assets in real
estate

Derive at least 75%
of income from real
estate rental income
or interest on
mortgages

Countries may specify

A minimum number of shareholders

A maximum share ownership by a single shareholder

A minimum number of properties/ maximum asset
concentration

A maximum level of non-rental income

A maximum amount of development

Limits on leverage and types of loans.

[LOS 3.a] Discuss types of publicly traded real estate securities

3.

Advantages of REITs

Liquidity

Ability to buy and sell shares of almost any amount on major exchanges.

Transparency

Readily available share prices and transaction histories.

Diversification of property holdings

By property type, geography, and underlying tenant credit.

High-quality portfolios

Many companies own high-quality assets in leading markets.

Active professional management

Most companies have strong management overseeing property management teams with economies of scale.

Potentially stable income

Well-occupied properties subject to long-term leases generate predictable property income.

Tax efficiency

REIT structures avoid CIT, only taxed on dividends received.

[LOS 3.a] Discuss types of publicly traded real estate securities

4.

Disadvantages of REITs

Lack of retained earnings

REITs are required to pay 90%+ of earnings to shareholders as dividends → REITs must access capital markets to fund growth.

Regulatory costs

REITs have to maintain a public corporate structure and comply with regulatory

Reduced portfolio diversification benefits

REITs pricing is determined by stock market movements and liquidity rather than only by underlying value.

Limited in types of assets owned

REITs are also constrained in the types of assets they own. Many REITs form **taxable REIT subsidiaries (TRS)**, which pay income taxes on earnings from non-REIT-qualifying activities.

[LOS 3.a] *Discuss* types of publicly traded real estate securities

Further reading: REOCs

- Greater flexibility in types of properties.
- Not subjected to same operating requirement (e.g. are not mandated to distribute income to shareholders, reinvesting net incomes into new acquisitions or renovating old properties for resale).
- Flexibility to make developments and capital improvements on their lands.
- Must pay standard corporate taxes that REITs are exempted from paying.

REITs vs. REOCs

REITs

- Exemption from taxation → Must pay higher dividends (90 - 100%), can only return less than 10% of net income to reinvest
- Must meet certain operating requirements to maintain status.
- Limited potential income growth

REOCs

- At liberty to set policies on how the net income is reinvested and distributed → Must pay standard corporate taxes that REITs are exempted
- Greater flexibility → higher potential for income growth

[LOS 3.a] Discuss types of publicly traded real estate securities**Knowledge check*****Example 1: Publicly Traded Real Estate Investments***

Which of the following assets requires the most expertise in real estate on the part of the investor?

- A. A REOC share
- B. An equity REIT share
- C. A direct investment in a single property

[LOS 3.a] *Discuss* types of publicly traded real estate securities



Knowledge check

Example 1: Publicly Traded Real Estate Investments

Which of the following assets requires the most expertise in real estate on the part of the investor?

- A. A REOC share
- B. An equity REIT share
- C. A direct investment in a single property

Answer:

Direct investment in a single property requires a high level of real estate expertise → **C is correct.**

Investment in publicly traded equity investments (in REITs or REOCs) requires much less expertise because the investors will have their property interests actively managed on their behalf by professional managers and have their business interests overseen and guided by boards of directors → **A and B are incorrect.**

[LOS 3.a] Discuss types of publicly traded real estate securities**Knowledge check*****Example 2: Publicly Traded Real Estate Investments***

Which of the following has the most operating and financial flexibility?

- A. A REOC share
- B. An equity REIT share
- C. A direct investment in a single property

[LOS 3.a] *Discuss* types of publicly traded real estate securities



Knowledge check

Example 2: Publicly Traded Real Estate Investments

Which of the following has the most operating and financial flexibility?

- A. A REOC share
- B. An equity REIT share
- C. A direct investment in a single property

Answer:

REOCs are free to invest in any kind of real estate or related activity without limitation with a wider range of capital structure choices.

→ Gives management the opportunity to create more value in development activity and in trading real estate and to retain as much of their income as they believe is appropriate.

→ **A is correct.**

REITs face restrictions on the amount of income and assets accounted for by activities other than collecting rent and interest payments.

→ **B is incorrect.**

Direct investment is less liquid and divisible than REOC and REIT shares, which limits the operational flexibility of such investment.

→ **C is incorrect.**

[LOS 3.b] *Justify* the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

1. Overview of valuing publicly traded real estate securities

Equity valuing approaches

Price multiple comparisons

Asset value estimates

Discounted cash flow

Valuing publicly traded real estate securities

Book value per share (BVPS)

based on reported accounting values

Net asset value per share (NAVPS)

based on market values for assets

Refers to depreciated real estate value rather than total shareholders' equity per share

Relevant market-based valuation measure for **valuing REITs and REOCs**

[LOS 3.b] Justify the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

1.

Overview of valuing publicly traded real estate securities

**NAVPS
should
include**

Assessments of any non-asset-based income streams

The value of non-real estate assets, including cash

Net of the value of any contingent liabilities

The value added by management of the REIT or REOC

**Shares
price**



NAVPS



**Potential
undervaluation**

**Shares
price**



NAVPS



**Potential
overvaluation**

REITs whose shares trade below NAVPS or have high leverage may have a more difficult time raising new capital → may limit long-term growth.

[LOS 3.b] *Justify* the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

2.
Accounting for Investment Properties

If companies use:

Cost model

Accounting values may be relevant for asset-based valuation

Fair value model

Accounting values are generally not relevant and **must be adjusted**

[LOS 3.b] Justify the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

2.

Accounting for Investment Properties

2.1. Under IFRS

Under IFRS, companies are allowed to value investment properties using either a cost model or a fair value model:

Cost model

Identical to the cost model used for PPE.

Disclosure:

- Depreciation method, useful life
- The fair value of investment property

Fair value model

Assumes all changes in fair value affect net income

Disclosure:

- How it determines fair value
- Reconciliation between the beginning and ending carrying amounts

A company must **consistently apply its chosen model** (cost or fair value) to all of its investment properties until it disposes of the property or changes its use such that it is no longer considered investment property.

[LOS 3.b] Justify the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

2.

Accounting for Investment Properties

2.2. Under USGAAP

Under US GAAP, most US real estate owners use the historical cost accounting model:

Value of an asset

= Original purchase price + Capital investment - Historical depreciation

Cost model distort the measure of economic income and asset value by:

- **Understating carrying values** on assets that are often appreciating in value because of inflation or other reasons.
- **Overstating depreciation** when companies use accelerated depreciation.

[LOS 3.b] Justify the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

3.

Net Asset Value per Share: Calculation

Definition

- The difference between a real estate company's current market value of its assets and liabilities, divided by the number of shares outstanding.
- Is a superior measure of a company's net worth compared with historical book value per share.

Net asset value per share (NAVPS)

$$= \frac{(\text{Market value of assets} - \text{Market value of liabilities})}{\text{Number of shares}}$$

If existing appraisals are unavailable or if analysts disagree with the assumptions or methodology of those appraisals, analysts will use net operating income (NOI) to calculate NAVPS (continue on next slide).

Net operating income (NOI)

$$= \text{Gross rental revenue} - \text{Estimated vacancy and collections loss} \\ - \text{Operating expenses}$$

[LOS 3.b] Justify the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

3.

Net Asset Value per Share: Calculation

NOI

Consider:

- Non-cash rent
- Impact of acquisitions
- NOI growth rate



Cap rate

Based on the prices of comparable recent transactions



Value of operating real estate

Value of operating real estate



Other tangible assets

Cash, account receivables, fixed asset
Exclude DTA, goodwill



Total debt + Other liabilities

Fair value of debt
Exclude DTL

Total shares outstanding



NAVPS

[LOS 3.b] *Justify* the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

3.

Net Asset Value per Share: Calculation


Illustration: Analyst Adjustments to REIT Financials to Obtain NAVPS

Last-12-month real estate NOI	\$270,432
Less: Non-cash rent	7,667
Plus: Adjustment for full impact of acquisitions ⁽¹⁾	4,534
Pro forma cash NOI for last 12 months	\$267,299
Plus: Next-12-month growth in NOI ⁽²⁾	\$4,009
Estimated next-12-month cash NOI	\$271,308
Assumed cap rate ⁽³⁾	7.00%
Estimated value of operating real estate	\$3,875,829
Plus: Cash and equivalents	65,554
Plus: Land held for future development	34,566
Plus: Accounts receivable	45,667
Plus: Prepaid/other assets ⁽⁴⁾	23,456
Estimated gross asset value	\$4,045,072

[LOS 3.b] Justify the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

3.

Net Asset Value per Share: Calculation



Illustration: Analyst Adjustments to REIT Financials to Obtain NAVPS

Estimated gross asset value	\$4,045,072
Less: Total debt	1,010,988
Less: Other liabilities	119,886
Net asset value	\$2,914,198
Shares outstanding	55,689
NAVPS	\$52.33

(1) An incremental 50% of the annual expected return on acquisitions that were completed midway through the previous year.

(2) Growth is estimated at 1.5%.

(3) The cap rate is based on recent comparable transactions in the property market.

(4) This figure does not include intangible assets.

[LOS 3.b] Justify the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

4.

Net Asset Value per Share: Application

NAVPS can be reasonably estimated when there are ample market transactions to provide property comparable.

Important Considerations in a NAV-Based Approach to Valuing REITs

Investors must understand the implications of **using a private market valuation tool on a publicly traded security** (examine how NAVs are calculated).

NAV calculation methods

Using the cap rate approach to valuing the NOI of property

Applying value per square foot (or unit) to a property

Using appraised values disclosed in the financial statements

[LOS 3.b] *Justify* the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

4.
Net Asset Value per Share: Application
NAV calculation methods
Invest in REITs

Cap rates and values per square foot are derived from observing transactions that have occurred in the marketplace.

Direct purchasers of properties

Cap rates and values price per square foot are achieved through a more detailed analysis and discounted cash flow process.

The discount rate used by a private owner/operator of commercial real estate could **differ** from the discount rate used by investors purchasing shares of REITs.

[LOS 3.b] *Justify* the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

4.

Net Asset Value per Share: Application

Premium or Discount to NAV

- Real estate stocks can trade at either premiums or discounts to NAV.
- The price-to-NAV ratio will vary by market, sector, outlook, and perceived quality of management and governance.

Private property investors

Consider the long-term prospects and valuation

REITs investors

Focus more on the near-term outlook for changes in income and asset value

Explain why share valuation may differ from NAV

[LOS 3.b] *Justify* the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

4.
Net Asset Value per Share: Application
Premium or Discount to NAV
NAV implicitly treats a company as an individual asset
Not consistent with a going-concern assumption, in fact:

Management teams have different track records and abilities

Assets can be purchased and sold

Capital market decisions can add or subtract value

Investor must consider how much value a management team can add to (or subtract from) current NAV

[LOS 3.b] *Justify* the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

4.**Net Asset Value per Share: Application****Premium or Discount to NAV**

NAV estimates can be subjective when property markets become illiquid

Large premium in good economy
when many strategic purchasers
transact

Large discount when there are
few buyers for the kind of
property

Complicate the process of estimating NAV with accuracy and confidence

[LOS 3.b] *Justify* the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income

4.

Net Asset Value per Share: Application

Further Observations on NAV

Common view: If REIT management is performing well in the sense of creating value, REITs and REOCs should trade at **premiums** to underlying NAVPS.

2 rationales

Investors in the stocks have liquidity on a day-to-day basis
→ Lower required return rate
→ Higher value

Competitive in the public markets
→ Attract above average management teams
→ Better real estate operating performance

Although NAV is by its nature an absolute valuation metric, in practice it is often **more useful as a relative valuation tool**

Purchasing the REIT trading at the smallest premium to NAV

Selling the REIT trading at the smallest discount to NAV

[LOS 3.c] Describe the use of funds from operations (FFO) and adjusted funds from operations (AFFO) in REIT valuation

1.

Funds from Operations; Adjusted Funds from Operations

REIT analysts and investors use two measures of operating performance

**Funds from operations (FFO)
(1.1)**

**Adjusted funds from operations
(AFFO) (1.2)**

1.1. Funds from operations (FFO)

FFO attempts to *approximate continuing operating performance*.

Funds from operations

$$\text{FFO} = (\text{Net income} + \text{Depreciation} + \text{Amortization}) - (\text{Net gains on sale of real property})$$

- **Depreciation** is added because it does not represent economic reality.
- **Adjusted for gains and losses from sales** of previously depreciated operating properties as they do not represent sustainable, normal income.

[LOS 3.c] Describe the use of funds from operations (FFO) and adjusted funds from operations (AFFO) in REIT valuation

1.

Funds from Operations; Adjusted Funds from Operations

1.2. Adjusted funds from operations (AFFO)

- AFFO is a refinement of FFO that is designed to be **a more accurate measure of current economic income**.
→ Superior to FFO: more reflective of a REIT's dividend-paying ability than FFO.
- However, it is open to more variation and error in estimation than FFO.

Adjusted funds from operations

$$\text{AFFO} = (\text{FFO} - \text{Non-cash rent} - \text{Recurring capex})$$

- Non-cash rent (***“straight-line rent adjustment”***) is the difference between ***straight-line rent*** (*the average contractual rent over a lease term – recorded as revenue*) and the cash rent actually paid.
- Recurring capex reflect costs to maintain the value of properties.

[LOS 3.c] Describe the use of funds from operations (FFO) and adjusted funds from operations (AFFO) in REIT valuation

1.

Funds from Operations; Adjusted Funds from Operations



Illustration

Calculation of FFO and AFFO for Office Equity REIT Inc. (SGD thousands, except per-share data)

A. Calculation of funds from operations (FFO)

Net income	160,638
Add: Depreciation and amortization	76,100
Add: (Gains)/losses from sale of depreciable real estate	25,000
Funds from operations	261,738
FFO per share (55,689 shares outstanding)	4.70

B. Calculation of adjusted funds from operations (AFFO)

Funds from operations	261,738
Less: Non-cash (straight-line) rent adjustment	21,103
Less: Recurring maintenance capital expenditures and leasing commissions	55,765
Adjusted funds from operations	184,870
AFFO per share (55,689 shares outstanding)	3.32

[LOS 3.c] Describe the use of funds from operations (FFO) and adjusted funds from operations (AFFO) in REIT valuation

2.

Relative Value Approach to Valuing REIT Stocks

For REITs, the relative value measures most frequently are **P/FFO** and **P/AFFO**. The ratio **EV/EBITDA** is used to a lesser extent.

Compare with other REITs shares

Compare with historical levels

Relative valuation approaches

P/FFO
P/AFFO
EV/EBITDA

Based on net income available to equity
→ represent levered income
→ **lower** for companies with **higher leverage**

Measures income before the leveraging effect of debt.
→ facilitate comparison

[LOS 3.c] *Describe* the use of funds from operations (FFO) and adjusted funds from operations (AFFO) in REIT valuation

2.

Relative Value Approach to Valuing REIT Stocks

Three main drivers that differentiate P/FFO, P/AFFO, and EV/EBITDA

Expectation for growth in FFO and AFFO

The higher the expected growth, the higher the multiple or relative valuation. Growth driven by business model, geography, other factors.

Risk of the underlying real estate

Cash flow volatility, quality, and age; market conditions; lease types; and submarket location

Capital structures and access to capital

As leverage increases, valuation multiples decrease, limits borrowing capacity and cause stock overhang.

[LOS 3.c] Describe the use of funds from operations (FFO) and adjusted funds from operations (AFFO) in REIT valuation

3. Advantages and Disadvantages

The key advantages and disadvantages of using P/FFO and P/AFFO multiples in the valuation of REITs and REOCs are as follows:

Advantages

Multiples are *widely accepted* in global stock markets and industries.

Portfolio managers can compare with *other alternatives*.

FFO estimates are *readily available* through market data providers.

Can be *used in conjunction* with expected growth and leverage levels to deepen analysis

Disadvantages

May not capture the intrinsic value of all real estate assets (ex: non-income-producing assets)

P/FFO does not adjust for the *impact of recurring* capital expenditures. P/AFFO is much subject to assumptions.

An increased level of *one-time items* has affected the income statement

[LOS 3.d] Calculate and interpret the value of a REIT share using the net asset value, relative value (price-to-FFO and price-to-AFFO), and discounted cash flow approach



Knowledge check

Example: Valuation

An analyst gathers the following information for two REITs:

	Price/NAV	Capitalization Rate Used in NAV
REIT A	100%	6%
REIT B	99%	8%

If the REITs have similar property portfolio values, interest expense, and corporate overhead, which REIT most likely has the higher price/FFO?

- A. REIT A
- B. REIT B
- C. They will have similar P/FFO because their ratios of price to NAV are almost identical.

[LOS 3.d] *Calculate and interpret* the value of a REIT share using the net asset value, relative value (price-to-FFO and price-to-AFFO), and discounted cash flow approach



Knowledge check

Answer:

If both companies have similar portfolio values as indicated in the text and by the similar P/NAV, then the company with the lower cap rate is more expensive, which results in lower FFO and hence a higher P/FFO.

→ **A is correct and B is incorrect.**

C is incorrect because it neglects the effect of the lower capitalization rate of REIT A.

Different valuation methods can yield different results.

→ An analyst should re-examine the assumptions made to investigate why the approaches are generating such different results.

[LOS 3.e] *Explain* advantages and disadvantages of investing in real estate through publicly traded securities compared to private vehicles

Both public and private real estate equity investments provide exposure to real estate properties.

Listed real estate's liquidity makes it easier to express a short-term view.

Private real estate investors can pursue a variety of strategies, such as merchant development, which is highly restricted for REITs.

[LOS 3.e] *Explain* advantages and disadvantages of investing in real estate through publicly traded securities compared to private vehicles

1.

Advantages and disadvantages of private real estate

Advantages	Disadvantages
Direct exposure to real estate fundamentals	Low liquidity
Stable returns/low volatility	Difficult-to-exit funds' redemption activity is high
Property performance drives returns	High fees and expenses
Low correlations with other asset classes	Fewer regulations to protect investors
Wide variety of strategies/few restrictions	Some managers focus on asset gathering over high profitability
Control (direct real estate and separate accounts)	High investment minimums and high net-worth requirement
Potential inflation hedge	Low transparency
Potential to earn illiquidity premium	High returns often derived from leverage

[LOS 3.e] Explain advantages and disadvantages of investing in real estate through publicly traded securities compared to private vehicles

2. Advantages and disadvantages of public real estate

Advantages	Disadvantages
Tracks real estate fundamentals over long term	High volatility (compared with private)
Liquidity	Equity market correlation is high in short term
Access to professional management	REIT structure limits possible activities
Potential inflation hedge	Stock prices may not reflect underlying property
Potential for strong alignment of interests	Dividends taxed at high current income tax rates
Tax-efficient structure avoids double taxation	Regulatory compliance costs are prohibitive for small companies

[LOS 3.e] Explain advantages and disadvantages of investing in real estate through publicly traded securities compared to private vehicles

2.

Advantages and disadvantages of public real estate

Advantages	Disadvantages
Potential for exposure to diversification	Poor governance can penalize stock performance
Low investment requirements	Companies with high leverage are penalized
Access to diverse sectors	
No special investor qualifications	
Low entry/exit costs	
Limited liability	
Greater regulation and investor protections	
High transparency	

[LOS 3.e] Explain advantages and disadvantages of investing in real estate through publicly traded securities compared to private vehicles

2. Advantages and disadvantages of public real estate

Compare the characteristics of private real estate and public real estate:

Characteristics	Private Real Estate (Direct Investment)	Public Real Estate (Equity REITs and REOCs)
Volatility	Low	High
Inflation hedge	High	High
Liquidity	Low	High
Diversification	High	Not as high (high correlation with stock market in short-term)
Tax	Tax benefits	Tax-efficient structure (REITs only) Dividends are taxed at high rate

[LOS 3.e] Explain advantages and disadvantages of investing in real estate through publicly traded securities compared to private vehicles

2.

Advantages and disadvantages of public real estate

Characteristics	Private Real Estate (Direct Investment)	Public Real Estate (Equity REITs and REOCs)
<i>Regulations</i>	Few	Greater
<i>Restrictions</i>	Few	REIT structure limits possible activities
<i>Fees & expenses</i>	High investment cost High operating expense	Low investment requirement
<i>Exit</i>	Difficult	Low cost
<i>Investor requirements</i>	High net-worth requirements	No special investor qualifications

MODULE 4:
HEDGE FUND STRATEGIES

LEARNING OUTCOMES

- 4.a.** Discuss how hedge fund strategies may be classified
- 4.b.** Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies
- 4.c.** Discuss investment characteristics, strategy implementation, and role in a portfolio of event-driven hedge fund strategies
- 4.d.** Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies
- 4.e.** Discuss investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies
- 4.f.** Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies
- 4.g.** Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies

LEARNING OUTCOMES

- 4.h.** Describe how factor models may be used to understand hedge fund risk exposures
- 4.i.** Evaluate the impact of an allocation to a hedge fund strategy in a traditional investment portfolio

[LOS 4.a] *Discuss* how hedge fund strategies may be classified

1.

Recall: Overview of Hedge Funds

Hedge fund is a private investment partnership and funds pool that uses a wide range of strategies and invests in complex products.



Generate high returns, either in an absolute sense or on a risk adjusted basis relative to its portfolio-level volatility.



Portfolio management

Creatively managed, involving one or more asset classes.

Often leveraged, generally takes both long and short positions and quite often uses derivatives

[LOS 4.a] Discuss how hedge fund strategies may be classified

2.

Characteristics of Hedge Funds

Legal/Regulatory overview

Overall regulatory constraints are far less than those for regulated investment vehicles.

Flexible mandates – Few investment constraints

Relatively unhindered in their trading and investment activities, such as short selling and use of derivatives.

Large investment universe

Has access to a wide range of assets outside the normal set of traditional investments.

Aggressive investment styles

Undertake strategies deemed too risky for traditional investment funds.

Relatively liberal use of leverage

Extensive use of leverage to make the return profile meaningful.

Hedge fund liquidity constraints

Liquidity constraints provide managers with greater ability to take and maintain existing positions.

Relatively high fee structures

Relatively high investment fees on investors, including management fees and incentive fees.

[LOS 4.a] *Discuss* how hedge fund strategies may be classified

3.

Classification of Hedge Fund Strategies

Equity-related

Focus primarily on the equity markets, risk profiles involve equity-oriented risk.

Event-driven

Focus on corporate events, primary risk is event risk(the possibility that an expected event will negatively affect a company or security.

Relative value

Focus on the relative valuation between two or more securities, often exposed to credit and liquidity risks

Opportunistic

Focus on a multi-asset opportunity set, the risks depend on the opportunity set involved

Specialist

Focus on special or niche opportunities that often require a specialized skill or knowledge of a specific market, exposed to unique risks

Multi-manager

Focus on building a portfolio of diversified hedge fund strategies

[LOS 4.a] Discuss how hedge fund strategies may be classified

3.

Classification of Hedge Fund Strategies

Hedge Fund Strategies by Category

Equity

- Long/short equity
- Dedicated short bias
- Equity market neutral

Event-driven

- Merger arbitrage
- Distressed securities

Relative value

- Fixed income arbitrage
- Convertible bond arbitrage

Opportunistic

- Global macro
- Managed futures

Specialist

- Volatility strategies
- Reinsurance strategies

Multi-manager

- Multi-strategy
- Fund-of-funds

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies

1.

Equity hedge fund strategies

Equity hedge fund strategies invest primarily in equity and equity-related instruments. The size and sign of equity market exposure often dictate the classification:

1

Long/Short equity

2

Dedicated short bias

3

Equity market-neutral

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies

2.

Long/short equity hedge fund strategies

Company A stock

Company B stock

Expect a rise in value

Expect a fall in value

Take long position

Take short position

Investment characteristics

- Combined long and short positions result in **portfolio beta** (i.e., market exposure) equal to the weighted sum of the positive and negative betas.
- Due to markets up trend over time, L/S funds will typically have a 40%-60% **net long exposure**.
- L/S funds provide returns comparable to those of a long-only fund but with **half the amount of standard deviation**.
- Leverage usage: variable – the more market-neutral the strategy approach, the more leverage is used to achieve a meaningful return profile.

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies

2.

Long/short equity hedge fund strategies

Strategy implementation

Requirement

Identification of underpriced and overpriced securities

Managers need to have thorough understanding of those stocks.

Sector-specific focus

Choosing securities from a particular industry that they are familiar with.

Role in a portfolio

Contribution

Derive alpha from long/short positions in stocks, while also benefitting from a moderate overall long exposure.

Investment consideration

Is the investment worthwhile given the potentially high fees?

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies

3.

Dedicated Short selling and Short-biased strategies

Dedicated short selling

Company X stock

Overpriced versus fundamental situation

Take short-only positions

Short-biased

Company X stock

Overpriced versus fundamental situation

Take short positions → Have short exposure

Balance with some long exposure

Activist short selling is a strategy whereby managers take a short position in a given security and then publicly present their research backing the thesis.

[LOS 4.b] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies

3.

Dedicated Short selling and Short-biased strategies

Investment characteristics

- Produce negative correlation with conventional securities
- Expected returns tend to be lower than other hedge funds
- Greater volatility than long short strategy due to unbalanced beta exposure
- Low use of leverage

Benefits

- Higher diversification due to lower correlation
- Highest return in the declining market

Drawbacks

- Unlimited downside risk
- High risk due to rising nature of market

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies

3.

Dedicated Short selling and Short-biased strategies

Strategy implementation

1 Bottom-up approach: Scanning the universe of potential sell targets.



*Flawed business
model*



*Unsustainable
leverage*



*Poor corporate
governance*



*Accounting
gimmickry*

2 Sell short: Borrowing the security and sell it at current market price.

3 Cover the position: Buying back the security and return to the lender.

Gain if able to buy back the security at lower price, loss if having to buy at higher price.

**Net short
exposure**

**Dedicated
short-sell**

Select stocks for a pure short exposure, typically 60-120% short.

**Short-
biased**

Take short exposure and take on some long exposure, often 30%-60% net short.

[LOS 4.b] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies

3.
Dedicated Short selling and Short-biased strategies
Role in a portfolio
Contribution

Produce returns that are uncorrelated with the return of conventional portfolio assets, provides a diversification benefit to a portfolio.

**Investment
consideration**

Expected returns for short strategies are relatively low.

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies

4.

Equity market neutral hedge fund strategies

Equity market neutral strategies seek to attain a near-zero overall exposure to the stock market.

Company A stock

Company B stock

Expect a rise in value

Expect a fall in value

Take long position

Take short position

Weighted betas of these positions should sum to zero

**Subset of EMN
strategies**

Pairs trading

Two stocks with similar characteristics but are respectively overvalued and undervalued.

Stub trading

Going long and short shares of a subsidiary and its parent company.

**Multi-class
trading**

Going long and short relatively mispriced share classes of the same firm.

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies

4.

Equity market neutral hedge fund strategies

Investment characteristics

- Generates return and alpha without beta risk
- Immune to movements in the overall market
- Returns tend to be modest, and offer significant diversification and low volatility
- Many different types of EMN managers exist, but many are purely quantitative managers
- As many beta risks are hedged away, it is acceptable for the use of **high leverage**

[LOS 4.b] *Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies*

4.

Equity market neutral hedge fund strategies

Strategy implementation

1 *Investment universe is evaluated*

Include only tradable securities with sufficient liquidity and adequate short-selling potential.

2 *Analyzing securities*

Analyzed for buy and sell opportunities using fundamental models or statistical and momentum-based models.

3 *Market risk neutrality constraints*

A portfolio is constructed with the market value-weighted beta is approximately zero.

4 *Availability and cost of leverage*

Consider the cost of leverage in terms of desired return profile and acceptable potential portfolio drawdown risk.



Takeaway: EMN manager's job is to actively manage the funds' exposure to remain neutral over time, while not allow the costs to overwhelm the security-selection alpha.

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies

4.

Equity market neutral hedge fund strategies

Role in a portfolio

Contribution

Attempt to produce alpha without taking market beta risk, particularly successful when markets are volatile and performing poorly.

Investment consideration

Costs of leverage must not overwhelm the alpha generated.

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies



Knowledge check

Question 1: Consider the following equity hedge fund strategies, the strategy that is *most likely* to apply relatively high levels of leverage is a(n):

- A. Equity market neutral strategy.
- B. Dedicated short strategy.
- C. Short-biased strategy.

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies



Knowledge check

Question 1: Consider the following equity hedge fund strategies, the strategy that is *most likely* to apply relatively high levels of leverage is a(n):

- A. Equity market neutral strategy.
- B. Dedicated short strategy.
- C. Short-biased strategy.

Answer: A

Recall: Leverage usage of equity-related strategies

Long/short equity	Dedicated short/short-biased	Equity market neutral
Variable leverage	Low leverage	High leverage

Equity market neutral strategy: As many beta risks are hedged away, it is acceptable for the use of high leverage.

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies



Knowledge check

Question 2: An equity-related hedge fund strategy with gross exposures of 80% long and 35% short is *most likely* to be classified as a:

- A. Dedicated short strategy.
- B. Short-biased strategy.
- C. Long/short equity strategy.

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies



Knowledge check

Question 2: An equity-related hedge fund strategy with gross exposures of 80% long and 35% short is *most likely* to be classified as a:

- A. Dedicated short strategy.
- B. Short-biased strategy.
- C. Long/short equity strategy.

Answer: C

Exposure given: “80% long and 35% short” → Net long exposure

Recall: Net exposure of strategies

Long/short equity	Dedicated short	Short-biased
40% - 60% net long	60% - 120% short	30% - 60% net short

The question is asking for net long exposure, thus A and B are incorrect.

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies



Knowledge check

Question 3: Relative to other hedge fund strategies, equity market neutral strategies are *most likely* to:

- A. Exhibit relatively modest return.
- B. Be vulnerable to periods of market weakness.
- C. Earn return from alpha and beta risk.

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies



Knowledge check

Answer: A

Recall: Equity market neutral strategies

Company A stock

Company B stock

Expect a rise in value

Expect a fall in value

Take long position

Take short position

Weighted betas of these positions should sum to zero

- Weighted betas sum to zero → No beta risk, C is incorrect.
- No beta risk or lack of market exposure make EMN strategies attractive in periods of market weakness → B is incorrect.
- Because portfolio is aimed to be market neutral, this strategy have relatively modest return profile → A is correct.

[LOS 4.c] Discuss investment characteristics, strategy implementation, and role in a portfolio of event-driven hedge fund strategies

1.

Event-driven hedge fund strategies

Event-driven hedge fund strategies take positions in corporate securities and derivatives that are attempting to profit from the outcome of corporate events.

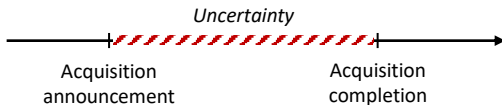
1
Merger arbitrage
2
Distressed securities

[LOS 4.c] Discuss investment characteristics, strategy implementation, and role in a portfolio of event-driven hedge fund strategies

2.

Merger arbitrage strategies

Merger arbitrage strategies attempt to earn a return from the uncertainty that exists in the market during the time between acquisition announcement and completion.



Uncertainty of an acquisition

**Announce-
ment**

Target's stock price rises, acquirer's stock price falls

Closing

Successful acquisition

Transaction fails

Prices move in the initial
directions

Price movements are
reversed

Long positions on target
gain

Long positions on target
incur losses

[LOS 4.c] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of event-driven hedge fund strategies

2.

Merger arbitrage strategies

Investment characteristics

- Correct predictions give significant profit, but incorrect predictions also give significant loss.
- Left-tail risk is higher.
- More liquid strategy comparing to other hedge fund strategies.
- Moderate to high leverage (3x to 5x) to generate meaningful target return levels.

[LOS 4.c] Discuss investment characteristics, strategy implementation, and role in a portfolio of event-driven hedge fund strategies

2.

Merger arbitrage strategies

Strategy implementation

Hedge fund manager's opinion

Successful deal

Failure in deal

**Long target stock
Short acquirer stock**

**Short target stock
Long acquirer stock**

Role in a portfolio

Contribution

Sharpe ratios tend to be high, usually produce relatively steady returns.

Investment consideration

There is significant left-tail risk.

[LOS 4.c] Discuss investment characteristics, strategy implementation, and role in a portfolio of event-driven hedge fund strategies

3.

Distressed securities strategies

Distressed securities strategies take positions in the securities of companies that are experiencing financial difficulties.



Too much leverage



Competition within sector



Accounting irregularities

Trades at greatly depressed prices

Hedge funds seek to own and exert control in the reorganization process

[LOS 4.c] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of event-driven hedge fund strategies

3.

Distressed securities strategies

Investment characteristics

- Higher return than other strategies but with higher variability.
- Usually long-biased.
- The lock-up periods for investors tend to be comparatively long (often allowing no redemptions for the first two years).
- Moderate to low use of leverage.

[LOS 4.c] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of event-driven hedge fund strategies

3.

Distressed securities strategies

Strategy implementation

- Distressed investing can take different forms:
 - Active: Building concentrated positions and placing representatives on the boards to turn the situation around.
 - Passive: Relying on others to bear the legal costs of reorganization.
- Requires a broad range of skills in order to navigate the various legal aspects of the strategy.

Role in a portfolio

Contribution

Returns are higher than other strategies though they are unpredictable and sensitive to decline market

Investment consideration

High level of illiquidity due to the nature of the assets.

[LOS 4.c] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of event-driven hedge fund strategies



Knowledge check

Question: An investment in distressed securities is *most likely* to be characterized by a:

- A. Long bias
- B. High level of liquidity
- C. Large amount of leverage

[LOS 4.b] Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies



Knowledge check

Question: An investment in distressed securities is *most likely* to be characterized by a:

- A. Long bias
- B. High level of liquidity
- C. Large amount of leverage

Answer: A

Recall: Distressed securities investment characteristics

- Usually long-biased
- The lock-up periods for investors tend to be comparatively long → High level of *illiquidity* → B is incorrect.
- Moderate to low use of leverage → C is incorrect.

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

1.

Relative value hedge fund strategies

Relative value hedge fund strategies exploit valuation differences between securities.

1

Fixed-income arbitrage

2

Convertible bond arbitrage

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

2.

Fixed-income arbitrage strategies

Fixed-income arbitrage strategies take advantage of temporary mispricing of fixed-income instruments.

Undervalued securities

Overvalued securities

Going long

Going short

Fixed-income instruments

Consumer debts

Bank loans

Corporate bonds

Sovereign bonds

Take positions in these securities in the belief that prices will revert toward their fair values

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

2.

Fixed-income arbitrage strategies

Investment characteristics

- Amount of profit that can be earned is somewhat limited, due to fixed-income being priced efficiently.
- Substantial leverage is often applied (4x – 15x).
- Liquidity depends on the strategy employed and the kinds of instruments used.

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

2.

Fixed-income arbitrage strategies

Strategy implementation

2.1. Yield curve trades

1 Anticipation

Hedge fund manager forms a view of how the shape of the yield curve will evolve over time.

2 Execution

Manager makes long and short investments in fixed-income instruments to profit from the anticipated yield curve steepening or flattening.

3 Profit

Manager will profit when the prices of the long securities rise and the short securities fall.

Sources of risk

Positions in securities of different firms

Liquidity risk, credit risk, interest rate risk

Positions in securities of same issuer

Interest rate movements

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

2.
Fixed-income arbitrage strategies

Strategy implementation

2.2. Carry trades

Strategy

- Short low-yielding security.
- Long high-yielding security.

**Source of
return**

- Yield differential
- Price changes

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

2.

Fixed-income arbitrage strategies

Role in a portfolio

Contribution

Payoff is similar to a short put option:

- Unfold as expected: positive carry return and return from spread narrowing.
- Unexpected widen spread: negative payoff

Investment consideration

Highly levered positions can cause modest price volatility to lead to a domino effect of margin calls and deleveraging.

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

3.

Convertible bond arbitrage strategies

Recall: Convertible bonds (*refer to Fixed Income topic*)

Convertible bonds

Represent an option-free bond and an embedded conversion option which gives bondholders the right to convert their debt into equity.

Convertible bonds can be viewed as a combination of straight bond plus a long equity call option.

Long convertible bond

=

Long the straight bond

+

Long call option on stock

Conversion price

The applicable share price at which the investor can convert to bonds into ordinary shares

Current share price < Conversion price

Current share price > Conversion price

Out-of-the-money call option

In-the-money call option

Convertible bond $\approx$ Straight bond

Convertible bond $\approx$ Underlying equity

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

3.

Convertible bond arbitrage strategies

Convertible bond arbitrage strategies profit from purchasing the implied volatility of convertible bonds – which is often underpriced.

Convertible bond arbitrageurs will take other positions in an attempt to hedge the delta and gamma risk of the convertible bond.

Recall: Delta and gamma (*refer to Derivatives topic*)

Delta

Delta measures the changes in *changes in option prices* as the asset price changes

Gamma

Gamma measures the rate of *change in delta* as the asset price changes.

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

3.**Convertible bond arbitrage strategies****Investment characteristics**

- Liquidity issues:
 - Naturally less-liquid securities because of small issue sizes and inherent complexities.
 - Availability and cost to borrow underlying equity for short selling.
- High use of leverage

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

3.
Convertible bond arbitrage strategies
Strategy implementation
Strategy

Buy relatively undervalued convertible bond and take short position in the relatively overvalued underlying stock of an amount equal to delta.

Outcomes

Current share price > Conversion price

Current share price < Conversion price

In-the-money call option

Out-of-the-money call option

Delta increases

Delta decreases

Short sell more stocks

Buy stocks to cover short position

The convertible arbitrage strategy will be profitable given sufficiently large stock price swings and proper period rebalancing.

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies

3.
Convertible bond arbitrage strategies
Role in a portfolio
Contribution

Performs well during periods of normal market conditions:

- Liquidity is available
- Volatility is moderate
- Convertible issuance is high

Investment consideration

May not perform well in periods of illiquidity or acute credit weakness.

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies



Knowledge check

Question: In implementing a convertible arbitrage strategy, a portfolio manager is *most likely* to take a position that is:

- A. Long convertible bonds and short equity.
- B. Long straight bonds and short convertible bonds.
- C. Long convertible bonds and short straight bonds.

[LOS 4.d] Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies



Knowledge check

Question: In implementing a convertible arbitrage strategy, a portfolio manager is *most likely* to take a position that is:

- A. Long convertible bonds and short equity.
- B. Long straight bonds and short convertible bonds.
- C. Long convertible bonds and short straight bonds.

Answer: A

Recall: Convertible bond strategy implementation

- Buy relatively undervalued convertible bond
 - Take short position in the relatively overvalued underlying stock of an amount equal to delta
- Long convertible bonds and short equity.

[LOS 4.e] Discuss investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies

1.

Opportunistic hedge fund strategies

Opportunistic hedge fund strategies seek to profit from investment opportunities across a wide range of markets and securities using a variety of techniques.

1

Global macro strategies

2

Managed futures

Asset
classes

Sectors and
regions

Macro
themes

Multi-asset
relationships

Global basis rather than focusing on individual security level

[LOS 4.e] Discuss investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies

1.

Opportunistic hedge fund strategies

Categorization

Types of analysis and approach

Technical

Fundamental

Implementation of trading decision

Discretionary

Systematic

Types of instruments and markets

Types of analysis and approach

Technical analysis

Uses past price changes to forecast future price movements.

Fundamental analysis

Analyze security prices, markets, sectors, and the relationships between markets, using economic data as input.

[LOS 4.e] Discuss investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies

1.

Opportunistic hedge fund strategies

Categorization

Types of analysis and approach

Technical

Fundamental

Implementation of trading decision

Discretionary

Systematic

Types of instruments and markets

Implementation of trading decision

Systematic implementation

Employ computer algorithms and rules to determine which trades to make.

Discretionary implementation

Managers use their instinct to determine when to trade.

[LOS 4.e] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies

2.

Global macro strategies

Global macro strategies attempt to make correct assessments and forecasts of various global economic variables (e.g., inflation, currency exchange rates, yield curves, central bank policies, economic health,...)

Investment characteristics

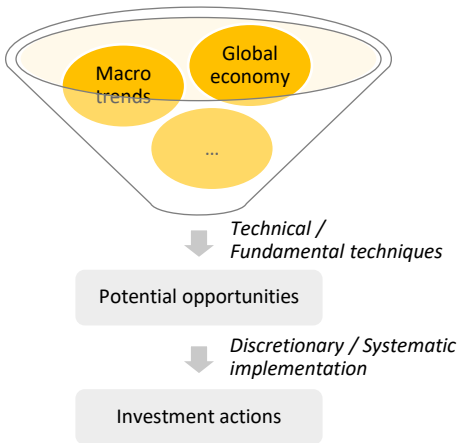
- Positions managers take can either be:
 - Directional: Long stocks that are anticipated to benefit from expected interest rate hikes, short stocks that will be disadvantaged.
 - Thematic: Buy stocks of firms that will benefit from forthcoming free trade deals.
- There is significant potential for unsuccessful investments when global economies do not behave as expected.
- High use of leverage (6x - 7x of fund assets).

[LOS 4.e] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies

2.

Global macro strategies

Strategy implementation



Global macro managers tend to use **discretionary approaches** more than do managed futures managers.

[LOS 4.e] Discuss investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies

2.
Global macro strategies
Role in a portfolio
Contribution

- Add alpha and portfolio diversification to a traditional asset portfolio.
- Attempt to anticipate changes before others → Contrarian tendency can make an allocation to this strategy advantageous.

Investment consideration

Although these strategies have delivered right-tail skewed returns, this behavior cannot always be relied upon.

[LOS 4.e] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies

3.

Managed futures strategies

Managed futures strategies take long and short positions in a variety of derivatives contracts.

Investment characteristics

- Do not buy and sell assets, but enter into derivatives contracts in order to gain the desired exposures.
- Great amount of leverage.
- Managed futures funds are extremely liquid, because futures contracts themselves are highly liquid.
- **Crowding:** many market participants pursue the same trades and use similar signals, resulting in a herding effect and execution slippage.

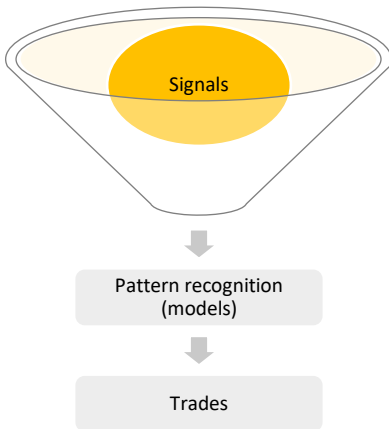
[LOS 4.e] Discuss investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies

3.
Managed futures strategies

Strategy implementation

Systematic implementation

Employ computer algorithms and rules to determine which trades to make.



[LOS 4.e] Discuss investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies

3.

Managed futures strategies

Strategy implementation

2.1. Time-series momentum

- Driven by past returns of the *individual assets*
- Strategy: long assets that are rising in price, short assets that are falling in price.

Long

VCB

MSN

VNM

Short

HPG

HDB

TLG

- Result: net long or net short depending on current price trend

2.2. Cross-sectional momentum

- Driven by performance of a *cross-section of assets*
- Strategy: long assets that are rising in price the most, short assets that are falling the most.

Long

VCB: 3.24%

MSN: 1.94%

VNM: 1.04%

Short

HPG: -0.18%

HDB: -1.23%

TLG: -2.45%

- Result: net zero or market-neutral position

[LOS 4.e] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies

3.**Managed futures strategies****Role in a portfolio****Contribution**

- Managed futures have very little correlation with traditional equity and fixed income assets
- Improve the total-risk adjusted return

[LOS 4.e] Discuss investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies



Knowledge check

Question: Considering global macro strategies and managed futures strategies, it would be *most accurate* to state that:

- A. Managed futures strategies use more discretionary approaches.
- B. Global macro strategies use more systematic approaches.
- C. Both strategies tend to use high leverage.

Answer: C

Recall: Implementation approaches:

- Discretionary approach: Managers use their instinct to determine when to trade → Frequently used in global macro strategies.
- Systematic approach: Employ computer algorithms and rules to determine which trades to make → Frequently used in managed futures strategies.

→ A and B are incorrect.

Global macro strategy

Managed futures strategy

High leverage: Notional amount up to 6x – 7x a fund's assets

→ C is correct.

[LOS 4.f] Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies

1.

Specialist hedge fund strategies

Specialist hedge fund strategies require highly specialized skill sets and knowledge of a particular market to pursue niche investment opportunities.

1

Volatility trading

2

Reinsurance/Life settlements

[LOS 4.f] Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies

2.

Volatility trading

Volatility trading hedge fund managers trade volatility-related assets globally, across countries and across asset classes, in order to **exploit perceived differences in volatility prices**.

Underpriced volatility

Overpriced volatility

Purchase

Sell

In the U.S. markets, the most common volatility futures are contracts on the **VIX index**, which tracks the 30-day implied volatility of the S&P 500 index. VIX contracts tend to be mean reverting because high volatility naturally tends to dissipate over time.

[LOS 4.f] Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies

2.
Volatility trading

Investment Characteristics

Position in volatility

Short position: Collect premiums, earn stable returns under average market conditions.

Long position: Exhibit positive convexity, which can be valuable as a hedge.

Liquidity

Varies with instruments:

- VIX index futures and options are very liquid
- OTC contracts are less liquid

Leverage

Can be applied easily

[LOS 4.f] Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies

2.

Volatility trading

Strategy implementation

Hedge fund managers wishing to pursue volatility trading have several options.

1 Construct various option strategies

Include only tradable securities with sufficient liquidity and adequate short-selling potential.

2 Make use of over-the-counter (OTC) options

Which are customized to meet the portfolio manager's specific needs.

Notes: This strategy is exposed with counterparty and liquidity risk.

3 Use futures on the VIX index

This is a very direct way to express a view on volatility, without the need for hedging. Too many investors also making it difficult to profit from that strategy.

4 Enter into an OTC volatility swap

These derivatives provide a relatively pure exposure to volatility.

[LOS 4.f] Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies

2.

Volatility trading

Role in a portfolio

Contribution

In a portfolio, a long volatility strategy can be a potent diversifier, because stock market volatility is highly negatively correlated with market returns.

Investment consideration

A premium must be paid to the volatility seller.

[LOS 4.f] Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies

3.

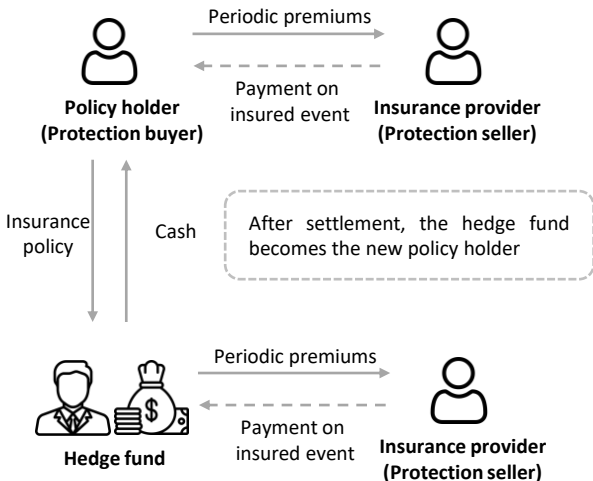
Reinsurance/Life Settlements

In a typical life settlement transaction, a hedge fund will buy (generally through a broker) insurance policies from a seller and become the new policy holder.

Before Transaction

Transaction

After Transaction



[LOS 4.f] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies

3.**Reinsurance/Life Settlements****Strategy implementation**

In selecting which insurance policies to invest in, a hedge fund will seek out policies with the following characteristics:

- The purchase price of the policy is low
- The on-going premium payments are low
- The insured person is likely to die relatively soon

[LOS 4.f] Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies

3.

Reinsurance/Life Settlements

Strategy implementation

1 Analyzing pools of life insurance contracts being offered for sale

Include only tradable securities with sufficient liquidity and adequate short-selling potential.

2 Pays a lump sum (via a broker) to the policyholder(s)

On finding the appropriate policy (or, more typically, a pool of policies), the hedge fund manager pays a lump sum (via a broker) to the policyholder(s), who transfers the right to the eventual policy benefit to the hedge fund.

3 Right transferred to the hedge fund

The hedge fund is then responsible for making ongoing premium payments on the policy in return for receiving the future death benefit.

This strategy is successful when the present value of the **future benefit** payment received by the hedge fund **exceeds** the **present value of intervening payments** made by the hedge fund.

[LOS 4.f] *Discuss* investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies

3.

Reinsurance/Life Settlements

Investment Characteristics

Strategies that involve investments in insurance contracts are illiquid, because insurance policies are somewhat difficult to sell after initiation.

Role in a portfolio

Contribution

In a portfolio, the risk inherent in these strategies is **almost entirely uncorrelated** with market risks and business cycles.

[LOS 4.f] Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies



Knowledge check

Question: A hedge fund is *most likely* to purchase a pool of life insurance policies that has high:

- A. surrender value.
- B. ongoing premium payments.
- C. likelihood of the insured person dying soon.

[LOS 4.f] Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies



Knowledge check

Question: A hedge fund is *most likely* to purchase a pool of life insurance policies that has high:

- A. surrender value.
- B. ongoing premium payments.
- C. likelihood of the insured person dying soon.

Answer: C

Recall: In implementing life settlement strategies, a hedge fund manager looks for policies with the following traits:

- Low surrender value being offered to the insured individual → A is incorrect.
- Low ongoing premium payments required of the investor → B is incorrect.
- High probability that the insured person will die sooner than predicted by actuarial methods → C is correct.

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies

1.

What Are Multi-Manager Strategies?

In the previous sections, multiple individual hedge fund strategies were introduced. However, in practice, most investors invest in **a range of hedge fund strategies for the sake of diversification**. There are three approaches to multi-manager strategies

1

Investing directly into different hedge funds

2

Fund-of-funds

3

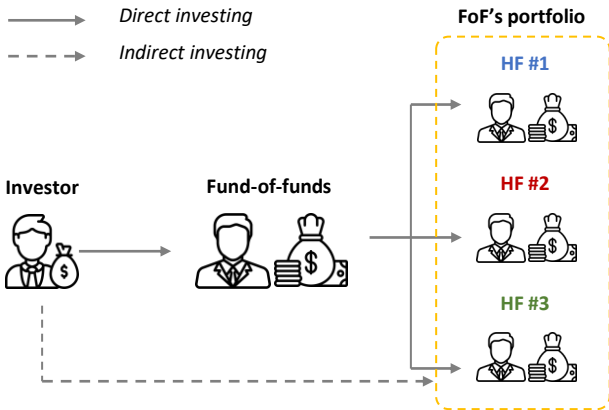
Multi-strategy funds

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies

2.

Fund-of-funds

Fund-of-funds (FoF) managers aggregate investors' capital and allocate it to a portfolio of separate, individual hedge funds following different, less correlated strategies.



[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies

2.

Fund-of-funds

Investment Characteristics

Advantages:

- Diversification across many hedge fund strategies.
- Expertise in individual manager selection.
- Strategic allocation and style allocation.
- Better liquidity relative to individual hedge funds.
- Access to certain closed hedge funds.

Disadvantages:

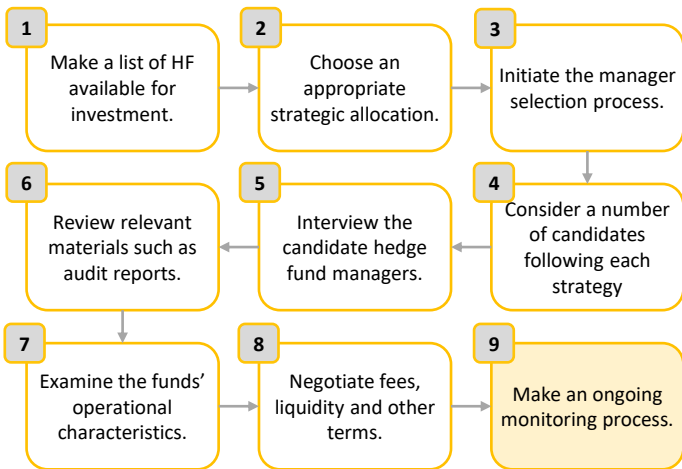
- A second layer of fees for the investor.
- Lack of transparency into individual hedge funds.
- No netting of performance fees.
- Additional principal–agent issues.

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies

2.

Fund-of-funds

Strategy implementation



[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies

2.
Fund-of-funds

Role in a portfolio

Contribution

- Greater diversification
- Steady returns
- Less concentrated exposure to risks
- Less volatility
- Less exposure to the downside risk of any individual fund manager

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies



Example: Net-of-Fee Returns

Question: Squaw Valley Fund of Funds (SVFOF) charges a 1% management fee and 10% incentive fee and invests an equal amount of its assets into two individual hedge funds: Pyrenees Fund (PF) and Ural Fund (UF), each charging a 2% management fee and a 20% incentive fee. For simplicity in answering the following questions, please ignore fee compounding and assume that all fees are paid at year-end.

1. If the managers of both PF and UF generate 20% gross annual returns, what is the net-of-fee return for an investor in SVFOF?

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies



Example: Net-of-Fee Returns

Answer:

1. Incentive fees are deducted only from gross gains net of management fees and expenses. Thus, the answer becomes:

Net of Fees Return for PF and UF Investor = $(20\% - 2\% - 3.6\%) = 14.4\%$,
where 3.6%

= $20\% \times (20\% - 2\%)$;

Net of Fees Return for SVFOF Investor = $(14.4\% - 1\% - 1.34\%) = 12.06\%$,
where 1.34%

= $10\% \times (14.4\% - 1\%)$.

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies



Example: Net-of-Fee Returns

Question: Squaw Valley Fund of Funds (SVFOF) charges a 1% management fee and 10% incentive fee and invests an equal amount of its assets into two individual hedge funds: Pyrenees Fund (PF) and Ural Fund (UF), each charging a 2% management fee and a 20% incentive fee. For simplicity in answering the following questions, please ignore fee compounding and assume that all fees are paid at year-end.

2. If PF's manager earns a gross return of 20% but UF's manager loses 5%, what is the net-of-fee return for an investor in SVFOF?

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies



Example: Net-of-Fee Returns

Answer:

1. Net of Fees Return for PF Investor = $(20\% - 2\% - 3.6\%) = 14.4\%$;

Net of Fees Return for UF Investor = $(-5\% - 2\% - 0\%) = -7.0\%$;

Gross Return for SVFOF Investor = $(0.5 \times 14.4\% + 0.5 \times -7.0\%) = 3.7\%$;

Net of Fees Return for SVFOF Investor = $(3.7\% - 1\% - 0.27\%) = 2.43\%$,

where 0.27%

$= 10\% \times (3.7\% - 1\%)$.

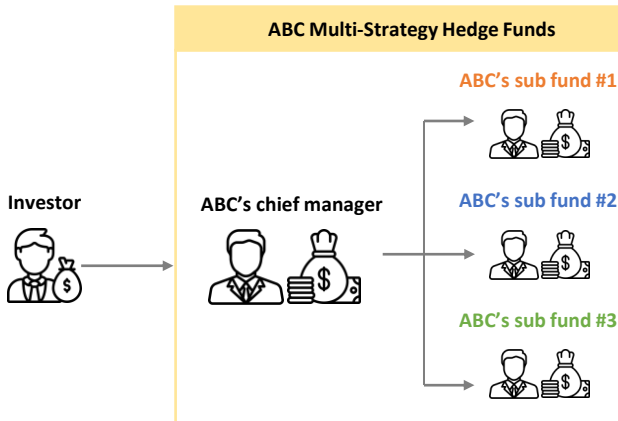
In conclusion, if both PF and UF managers generate gross returns of 20%, then the net-of-fee return for SVFOF's investor is 12.06%, with fees taking up 39.7% of the total gross investment return $[(2\% + 3.6\% + 1\% + 1.34\%)/20\% = 39.7\%]$ and the remainder going to the SVFOF investor.

But, if PF's manager earns a 20% gross return and UF's manager loses 5%, then the net-of-fee return for the SVFOF investor is a meager 2.43%. In this case, most (67.6%) of the original gross return of 7.5% $[= 20\% \times 0.50 + (-5\% \times 0.50)]$ goes to PF, UF, and SVFOF managers as fees. Note that $\{[0.50 \times (2\% + 3.6\% + 2\% + 0\%)] + (1\% + 0.27\%)\}/7.5\%$ equals 67.6%. This is an example of fee netting risk that comes with investing in FoFs.

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies

3.**Multi-Strategy Hedge Funds**

Multi-strategy hedge funds combine multiple hedge fund strategies to produce steady, low-volatility returns. **Sub-funds in multi-strategy hedge funds run by the same organization.**



[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies

3.**Multi-Strategy Hedge Funds****Investment Characteristics**

- Produce steady returns and low volatility.
- Operational risks of a multi-strategy fund are not well-diversified as they are with an FoF (sub funds perform under the same roof.)
- Can make quick tactical allocations.
- More attractive fees than FoF.
- Limit investor liquidity using redemption periods and initial lock-ups.

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies

3.

Multi-Strategy Hedge Funds

Strategy implementation

1 Invest in a range of individual hedge fund strategies

Multi-strategy funds can access is a function of the portfolio management skills available within the particular multi-strategy fund.

2 Engage in strategic and tactical allocations to hedge fund strategies.

Managers are more likely to be well informed about when to tactically reallocate to a particular strategy and more capable of shifting capital between strategies quickly.

3 Correlations between the different strategies run by the teams

Multi-strategy funds may also be less willing to exit strategies in which core expertise is in-house.

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies

3.**Multi-Strategy Hedge Funds****Role in a portfolio****Contribution**

Multi-strategy funds are intended to improve an investment portfolio by adding diversification and steady, low-volatility returns.

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies



Knowledge check

Question: Compared to a multi-strategy fund, a fund-of-funds is most likely to present an investor with higher:

- A. transparency.
- B. netting risks.
- C. leverage.

[LOS 4.g] Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies



Knowledge check

Question: Compared to a multi-strategy fund, a fund-of-funds is most likely to present an investor with higher:

- A. transparency.
- B. netting risks.
- C. leverage.

Answer: A

Recall: Compared to multi-strategy funds, funds-of-funds offer an investor less transparency and higher netting risk.

- Funds-of-funds offer an investor less transparency → A is incorrect.
- Funds-of-funds offer an investor higher netting risk → B is correct.
- Multi-strategy funds exhibit higher variance due to less diverse strategies and use comparatively higher leverage. → C is incorrect.

[LOS 4.h] Describe how factor models may be used to understand hedge fund risk exposures

1. Factor Models

1.1. Conditional Factor Risk Model

By conditional, we mean a model that takes into account that a fund may behave one way during normal market conditions, but perform differently during a period of market turbulence

Conditional Factor Risk Model

$$\begin{aligned}
 (\text{Return on HF}_i)_t = & \alpha_i + \beta_{i,1}(\text{Factor 1})_t + \beta_{i,2}(\text{Factor 2})_t + \dots \\
 & + \beta_{i,K}(\text{Factor 1})_K + D_t\beta_{i,1}(\text{Factor 1})_t + D_t\beta_{i,2}(\text{Factor 2})_t + \dots \\
 & + D_t\beta_{i,K}(\text{Factor 1})_K + (\text{error})_{i,t}
 \end{aligned}$$

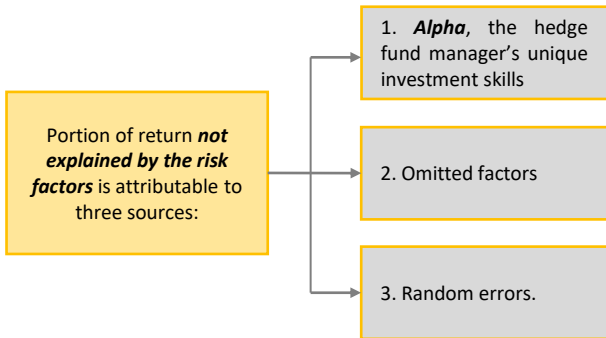
- **(Return on HF_i)_t**: return of hedge fund i in period t;
- **β_{i,1}(Factor 1)_t**: exposure to risk factor 1 (up to risk factor K) for hedge fund i in period t during normal times;
- **D_tβ_{i,1}(Factor 1)_t**: incremental exposure to risk factor 1 (up to risk factor K) for hedge fund i in period t during financial crisis periods, where D_t is a dummy variable that equals 1 during financial crisis periods (i.e., June 2007 to February 2009) and 0 otherwise;
- **α_i**: intercept for hedge fund i;
- **(error)_{i,t}**: random error with zero mean and standard deviation of σ_i.

[LOS 4.h] Describe how factor models may be used to understand hedge fund risk exposures

1. Factor Models

1.1. Conditional Factor Risk Model

Each factor beta represents the expected change in hedge fund returns for a one-unit increase in the specific risk factor, holding all other factors (independent variables) constant.



[LOS 4.h] Describe how factor models may be used to understand hedge fund risk exposures

1. Factor Models

1.2. Building Hedge Fund Risk Factor Models

Following Hasanhodzic and Lo (2007) and practice, the model starts with the following six factors:

Equity risk (SNP500)

Monthly total return of the S&P 500 Index, including dividends.

Interest rate risk (BOND)

Monthly return of the Bloomberg Barclays Corporate AA Intermediate Bond Index.

Currency risk (USD)

Monthly return of the US Dollar Index

Commodity risk (CMDTY)

Monthly total return of the Goldman Sachs Commodity Index (GSCI).

Credit risk (CREDIT)

Difference between monthly seasoned Baa and Aaa corporate bond yields provided by Moody's.

Volatility risk (VIX)

first-difference of the end-of-month value of the CBOE Volatility Index (VIX).

[LOS 4.h] Describe how factor models may be used to understand hedge fund risk exposures

1. Factor Models

Practical steps for building hedge fund risk factor models

1 *Identify potentially important risk factors*

Include only tradable securities with sufficient liquidity and adequate short-selling potential.

2 *Calculate pairwise correlations across all risk factors.*

If two-state conditional models are used, calculate correlations across all risk factors for both states.

3 *Regress the return series in case of highly correlated risk factors*

For highly correlated risk factors A and B, regress the return series of interest (e.g., hedge fund returns) on all risk factors excluding factor A.

4 *Repeat step 3 for all other highly correlated factor pairs*

with the aim of eliminating the least useful (in terms of explanatory power) factors and thereby avoiding multi-collinearity issues.

[LOS 4.h] Describe how factor models may be used to understand hedge fund risk exposures

1. Factor Models

Practical steps for building hedge fund risk factor models

- This stepwise regression process is used to create linear conditional factor models that **avoid multicollinearity problems**.
- After the process, the Hasanhodzic and Lo (2007)'s models' **six factors is reduced to four factors** for measuring risk exposure:

Normal Period

Equity risk (SNP500)

~~Interest rate risk (BOND)~~

Currency risk (USD)

~~Commodity risk (COMDTY)~~

Credit risk (CREDIT)

Volatility risk (VIX)

Crisis Period

Equity risk (DSNP500)

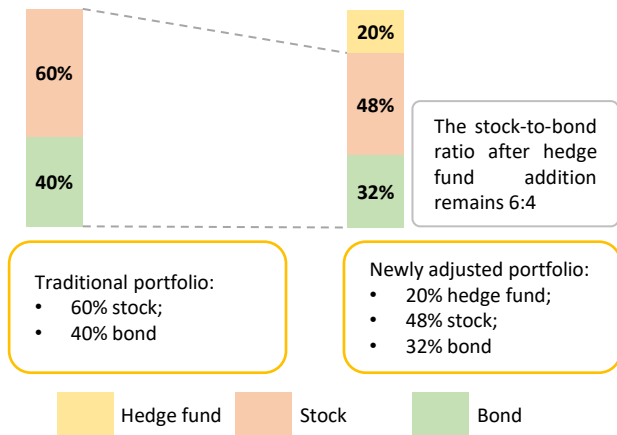
Currency risk (USD)

Credit risk (DCREDIT)

Volatility risk (DVIX)

[LOS 4.i] Evaluate the impact of an allocation to a hedge fund strategy in a traditional investment portfolio

This section examines the return and risk contributions of the hedge fund strategies previously covered when added to a traditional 60% stock/40% bond investment portfolio.



[LOS 4.i] Evaluate the impact of an allocation to a hedge fund strategy in a traditional investment portfolio

1.

Portfolio Contribution of Hedge Fund Strategies

1.1. Performance Contribution to a 60/40 Portfolio

***Effects when
strategies
added to
portfolio***

- Total portfolio standard deviation decreases.
- **Sharpe ratio** increases.
- **Sortino ratio** increases.
- Maximum drawdown decreases (in approximately one-third of portfolios).

Hedge fund strategies generally increase risk-adjusted return and provide diversification to a traditional portfolio of stocks and bonds.

[LOS 4.i] Evaluate the impact of an allocation to a hedge fund strategy in a traditional investment portfolio

1. Portfolio Contribution of Hedge Fund Strategies

1.2. Risk-Adjusted Performance

When 20% allocations to various hedge fund strategies are added to the traditional stock/bond portfolio, these targets are achieved via allocations to the following strategies:

High Sharpe ratios:

- Systematic futures.
- Distressed securities.
- Fixed-income arbitrage.
- Global macro.
- Equity market neutral.

High Sortino ratios:

- Equity market neutral.
- Systematic futures.
- Long/short equity.
- Event driven.

Generating superior risk-adjusted performance

- Systematic futures.
- Equity market neutral.
- Global macro.
- Event-driven hedge fund strategies.

Strategies **do not** significantly enhance risk-adjusted performance:

- Fund-of-funds.
- Multi-strategy.

[LOS 4.i] Evaluate the impact of an allocation to a hedge fund strategy in a traditional investment portfolio

2.

Risk Metrics

One key reason for investors to allocate a portion of their portfolio to hedge fund strategies is to reduce risk. We here examine the effect of risk reduction achieved by including HF strategies in traditional portfolios.

1.1. Standard Deviation

These targets are achieved via allocations to the following strategies:

Lowest standard deviations of returns for the overall portfolio:

- Dedicated short-biased.
- Bear market neutral.

Notably low standard deviations:

- Systematic futures.
- FoF: macro/systematic.
- Equity market neutral.

Little positive impact on reducing standard deviations of the overall portfolio include:

Systematic futures.

- Event-driven: distressed securities.
- Relative value: convertible arbitrage.

[LOS 4.i] Evaluate the impact of an allocation to a hedge fund strategy in a traditional investment portfolio

2.

Risk Metrics

1.2. Drawdown

Definition

The peak-to-trough decline for a portfolio, generally quoted as the percentage drop between a point of high portfolio value and the subsequent trough.

The hedge fund strategies that produced the smallest maximum drawdowns are various opportunistic strategies:

- Global macro.
- Systematic futures.
- Merger arbitrage.
- Equity market neutral.

Other hedge fund strategies did little to mitigate the traditional portfolio's maximum drawdown:

- Long/short equity.
- Event-driven: distressed securities.
- Relative value: convertible arbitrage.